



Sustainable cement and concrete technologies: a review of materials and processes for carbon reduction

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Abstract

This literature review investigates sustainable cement and concrete technologies with a focus on reducing carbon emissions. Key areas of exploration include Alternative Cementitious Materials (ACMs), Supplementary Cementitious Materials (SCMs), and innovative alternative clinkers. The review also examines low-carbon concrete mixtures using performance-based assessments and artificial intelligence techniques to optimize formulations, alongside energy-efficient production processes. By synthesizing current research and advancements, this paper highlights the critical role of sustainable practices in transforming the cement and concrete industries to achieve substantial carbon reductions and enhance environmental sustainability.

Keywords Cement · Carbon footprint · Cementitious materials · Alternative clinkers · Sustainability

Introduction

Reducing carbon emissions in the construction sector is essential for achieving carbon neutrality. The architecture, engineering, and construction industries collectively contribute about 40% of global carbon emissions, making it crucial to lower these emissions to combat climate change. Currently, the production of cement and steel is responsible for approximately 7% of annual carbon emissions. Most cement is used to create concrete or mortar, and about 52% of steel is utilized in reinforced concrete structures. Consequently, the construction of concrete buildings significantly contributes to the rise in carbon emissions. On a global scale, the annual consumption of reinforced concrete has reached over 10 billion tons.

In recent years, the growing emphasis on sustainability by both the engineering community and the general public has increased pressure on the construction industry. Building materials and construction are major contributors

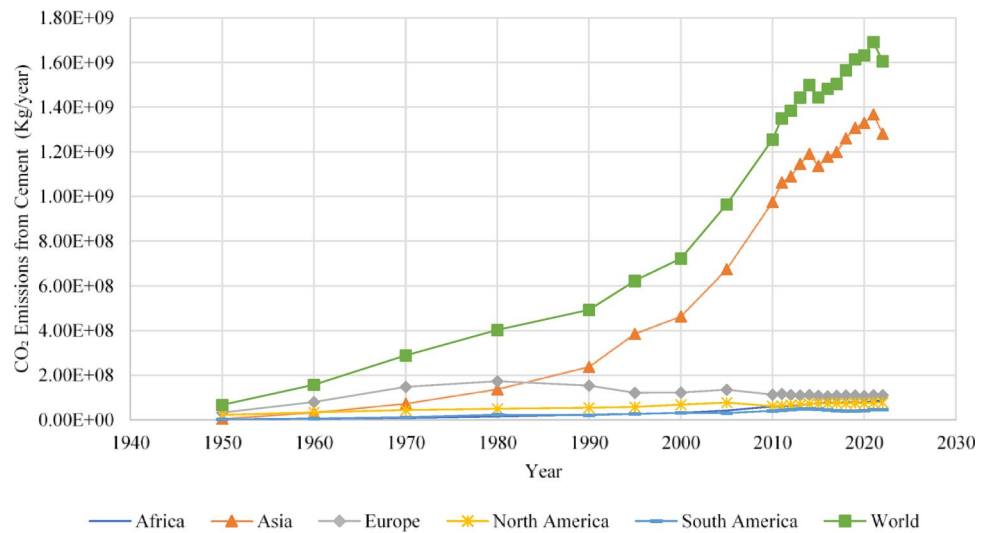
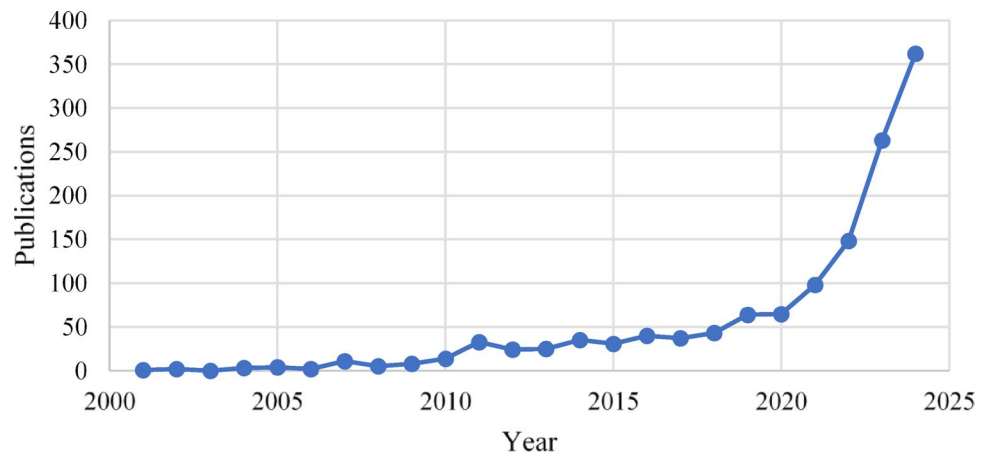
to global greenhouse gas (GHG) emissions, with concrete being the most consumed product after water and a significant source of CO₂ emissions each year. The GHG emissions from concrete stem from various stages, including material extraction and processing, transportation of raw materials and finished products, cement production, concrete mixing and production, and on-site construction activities. Although each stage has relatively low emissions, the cement used in concrete production is the primary culprit, contributing to 8% of the industry's total GHG emissions. Cement is indispensable for producing concrete due to its role as a binder that forms a solid material capable of sustaining loads. Ordinary Portland cement (PC) has been a crucial ingredient in concrete for over 200 years [1]. The process of cement manufacturing is accounting for more than 70% of GHG emissions in the concrete industry [2] and U.S. cement manufacturing accounts for less than 1.5% of all U.S. human-generated CO₂ emissions [3]. As shown in Fig. 1, the worldwide annual emissions of cement was reported as 1.61 billion ton in 2022 [4].

The cement production process consists of heating limestone, clay, and other materials in a kiln. The chemical reaction from the mixture and the energy used to fire the material are two sources of cement CO₂ emission. During the calcination of limestone (calcium carbonate) clinker is produced, which is the main ingredient in cement. This chemical reaction releases CO₂ as shown in Eq. (1).

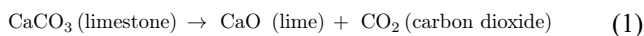
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Fig. 1 CO₂ emissions from cement [4]**Fig. 2** Journal publications on low-carbon concrete indexed in scopus

Furthermore, significant amounts of energy are required to heat the kiln to high temperatures (around 1450°C) needed for calcination. The fuel combustion for this heating process, often involving fossil fuels such as coal, natural gas, or oil, also generates substantial CO₂ emissions [5].

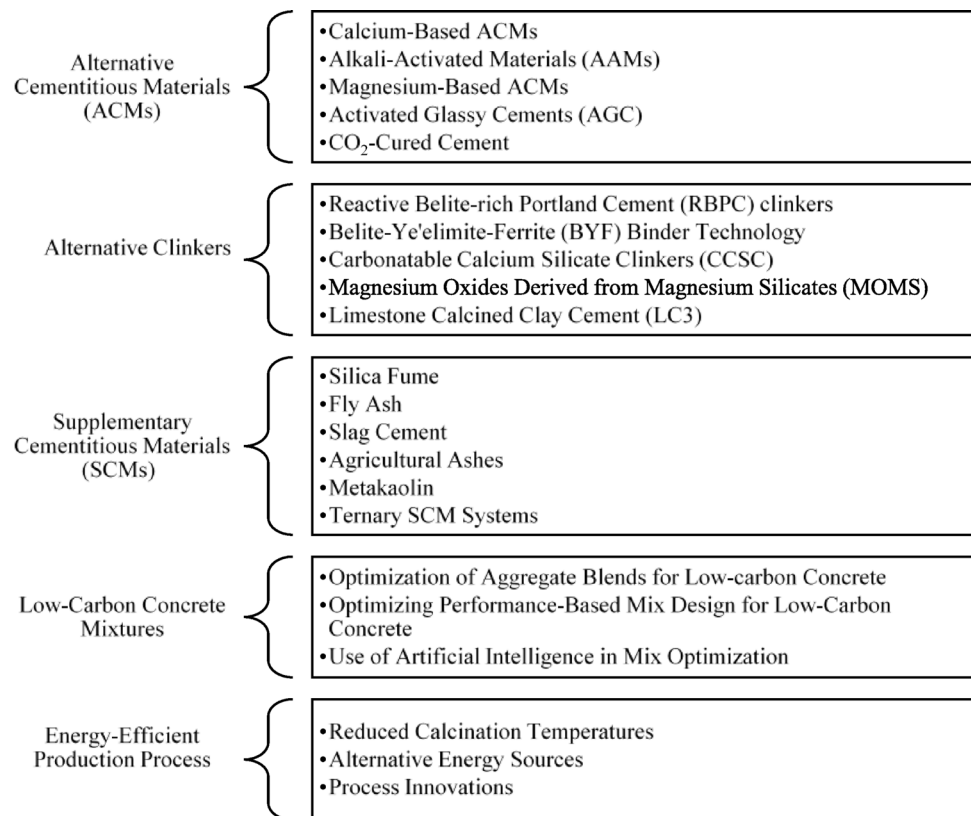


Carbon dioxide (CO₂), nitrous oxide (N₂O), and methane (CH₄) are the key GHGs associated with climate change. CO₂ is the most significant in terms of volume, accounting for the majority of emissions from cement production [6]. CO₂ emissions contribute significantly to global warming and climate change, which can have severe impacts on humans and the environment. These emissions act like a blanket in the atmosphere, trapping heat and warming the Earth [7]. The production of Portland cement has been found to emit approximately 0.8–1 tonne of CO₂ per tonne of cement produced [8]. Cement production alone accounts for over 7% of annual anthropogenic GHG emissions [9]. The process-based Life Cycle Assessment (LCA) approach

is employed to evaluate carbon emissions during the material production stage [10, 11].

In response to the pressing environmental challenges posed by traditional concrete production, researchers are intensifying efforts to reduce the carbon footprint of concrete and innovate carbon-neutral concrete solutions. This growing interest is evidenced by the rising number of academic publications featuring abstracts with the term “low-carbon concrete,” highlighting the global emphasis on sustainable construction practices. Figure 2 illustrates this trend, showcasing the increasing recognition and prioritization of low-carbon concrete in the academic and engineering communities.

The overarching goal of this review is to assess a range of strategies aimed at reducing the carbon emissions associated with concrete construction. These strategies include the use of alternative cementitious materials (ACMs), supplementary cementitious materials (SCMs), low-carbon concrete mixtures, and energy-efficient production processes, as illustrated in Fig. 3. For each approach, the review delves into the foundational principles of the technique and

Fig. 3 Low carbon cement and concrete techniques

highlights recent advancements and innovations that further enhance its potential for sustainable construction practices.

Alternative cementitious materials (ACMs)

This section explores various alternative cementitious materials (ACMs) that offer lower CO₂ emissions compared to Portland cement, highlighting their potential to reduce the environmental impact of concrete production. It begins with calcium-based ACMs, which are closely related to traditional cements but modified to lower carbon footprints through reduced clinker content or alternative raw materials. Following this, alkali-activated materials (AAMs) represent a distinct class of binders activated by alkaline solutions, offering unique chemical mechanisms and improved sustainability profiles. Magnesium-based ACMs then provide further alternatives, often featuring rapid carbonation processes that can capture CO₂ during curing, thus actively reducing emissions. Subsequently, activated glassy cements (AGC) introduce innovative compositions that leverage recycled glass and other waste materials to create efficient, low-carbon binders. Finally, CO₂-cured cements harness carbon dioxide curing techniques to enhance strength and durability while simultaneously sequestering CO₂, presenting a promising approach for both structural performance and carbon mitigation.

Calcium-based ACMs

Calcium aluminate cements (CACs), a calcium-based ACM, typically contain a high alumina content, ranging from 40 to 80%, along with a relatively low calcium oxide content [12]. The primary components of CACs include monocalcium aluminate (CA), with possible inclusion of C₁₂A₇ and/or CA₂. Small quantities of silica may be present as C₂S and/or C₂AS (gehlenite), while ferrite may appear in the form of a C₄AF solid solution with a variable A/F ratio. CACs are primarily utilized in refractory applications and specialized building chemistry, including floor screeds and rapid-hardening mortars [13].

CACs can withstand temperatures of 1600–2000 °C, making them suitable for high-heat and refractory applications [14]. They demonstrate high resistance to chloride-induced corrosion, with sulfate attack minimized due to the absence of Ca(OH)₂; they inhibit steel corrosion at a higher chloride threshold (~3% by weight of binder). This enhances their performance in harsh environments. CACs also achieve rapid early strength, reaching compressive strengths of 40 to 50 MPa within a few days, though strength may decline after 28 days due to “conversion,” where metastable hydrates transform into stable forms, increasing porosity and reducing strength. In chemically aggressive environments, including seawater and acidic conditions, CAC demonstrates exceptional durability, significantly extending

service life through its corrosion-resistant properties [15]. Furthermore, its lower calcination temperature contributes to a reduction in CO₂ emissions by 5–20% compared to Ordinary Portland Cement (OPC), thus offering more sustainable alternatives through clinker replacement [16].

Despite their benefits, the broader adoption of CACs faces challenges. The phenomenon of conversion compromises long-term strength, and the higher production costs—linked to the limited availability of bauxite, the primary alumina source—hinder wider use. Furthermore, rapid-setting mixtures can limit in-situ casting, and the brittleness and poor tensile properties restrict certain applications [13].

Calcium sulfoaluminate (CSA) cements contain ye'elimite (C₄A₃S) as a major constituent (30–70%). Due to their low pH, low porosity, and the ability of ettringite and AFM phases to bind heavy metals, CSAs and their blends with Portland cement are of interest in the field of hazardous waste encapsulation. In addition, CSA cement can significantly enhance the compressive strength, flexural strength, and durability of concrete. Replacing 20–30% of Portland cement with CSA has been shown to increase compressive strength by up to 30% [17]. It also exhibits lower drying shrinkage and permeability [18] and enhanced resistance to sulfate and chloride attack [19]. CSA cement provides fire resistance, especially in vermiculite mortars, making it suitable for fireproofing [20, 21]. It exhibits excellent resistance to sulfate and biogenic acid attack, outperforming ordinary Portland cement (OPC) in marine, sewer, and generally acidic environments [22–25]. CSA cement achieves rapid strength gain (e.g., 40 MPa in 6 h), enabling cold curing and accelerated construction [26, 27]. Additionally, CSA can reduce CO₂ emissions by 30–40% and lower embodied

carbon by 20–30%, compared to traditional Portland cement [28–31].

Super sulfated cements (SSCs) are clinker-free (or nearly clinker-free) binders typically composed of blast furnace slag, calcium sulfate, and a small amount of alkaline activator. Basic slags with high CaO and MgO content, and at least 14–15% Al₂O₃, are ideal for producing SSCs. SSCs are gaining attention in industrial applications, driven by their low heat of hydration, durability in chemically aggressive environments, and excellent sulfate resistance. SSCs offers good fire and heat resistance, making it suitable for mass concrete [32, 33]. It provides excellent resistance to sulfate and composite acid attacks, outperforming OPC in sulfate-rich environments due to its low Ca(OH)₂ content, enhancing durability [32, 34]. While strength development is slower, it achieves comparable strength (~35 MPa at 28 days, ~50 MPa at 90 days) [35–37]. This renewed interest is further evidenced by the recent introduction of new SSC products and the establishment of a European standard for super sulfated cements [38]. Using SSC as a binder has been shown to significantly reduce CO₂ emissions compared to traditional Portland cement. Studies report that SSC can decrease carbon dioxide emissions by at least 49%, with some research indicating potential reductions as high as 95% [39, 40].

Calcium Sulfoaluminate Belite (CSAB) cements, also known as sulfoaluminate cements or ye'elimite cements, are composed of a complex mixture of minerals, primarily calcium sulfoaluminate (also known as ye'elimite, 3CaO·3Al₂O₃·CaSO₄) and belite (2CaO·SiO₂). The CSA exhibits several advantages, including rapid hardening and early strength development, minimal shrinkage, exceptional resistance to freezing, and strong erosion resistance [41, 42]. However, the rapid strength development and stable hydration products such as ettringite and carboaluminate phases suggest that CSAB cements may perform well under thermal stress, but direct evidence on fire resistance is lacking [43].

An intriguing aspect of CSAB cement hydration is its potential for carbon dioxide capture and storage, which could further reduce its overall carbon footprint. While Portland cement production typically releases around 0.8–0.9 tons of CO₂ per ton of cement, CSAB can achieve reductions of up to 30–50%, depending on the specific mix and production conditions [13, 44].

Advancement in calcium-based ACMs

Recent research has focused on the performance, sustainability, and innovative applications of various ACMs. These advancements span several areas, as presented in Fig. 4.

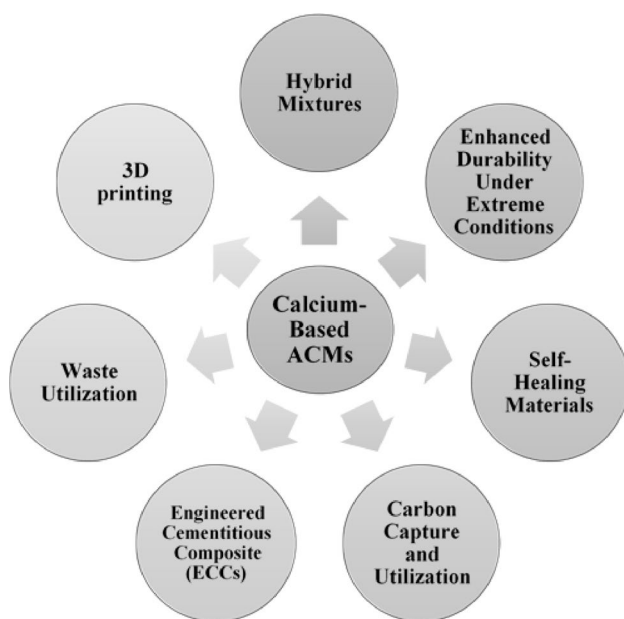


Fig. 4 Area of advancement for calcium-based ACMs

Hybrid mixtures: The development of hybrid binders that combine traditional cementitious materials with emerging alternatives is gaining traction. For example, Abolhasani, Nazarpour and Dehestani [45] investigated the influence of silica variation in CAC concrete, using rice husk ash (RHA) for its high silica content. They observed enhanced fracture properties at a 5% RHA substitution with a water-to-cement ratio (W/C) of 0.5. In another study, Jeong, Hargis, Chun and Moon [43] examined the effect of calcium carbonate (CC) fineness on the material properties of CSAB cements, finding that CC powder accelerated hydration, with finer CC particles having a greater effect due to more nucleation sites. Although finer CC particles compensated for the dilution effect of clinker, improving hydration, strength, and reducing pore volume, coarser CC particles diluted the clinker phases, resulting in reduced strength. Zhu, Yue, Aslam, Bai, Zhu and Wei [46] demonstrated that incorporating carbon nanotubes into CSAB could significantly enhance both early- and long-term strength, making it suitable for high-performance applications. Additionally, using thermally treated reservoir sediments in CSAB production promotes sustainability by repurposing waste materials, reducing reliance on virgin resources, and lowering the carbon footprint of construction projects. In another study, Wang, Sun, Tao and Pan [47] examined the role of silica fume (SF) in geopolymer-based UHPC, noting that SF's influence on compressive strength varied with CAC content, showing an optimal increase with steel fiber addition at 20% CAC but reduced strength at 10% CAC with higher SF content.

Enhanced durability under extreme conditions: Research on the durability of CAC and CSA cements under extreme conditions has also advanced significantly. Studies have shown that these materials offer superior performance in high-temperature applications due to the formation of thermally stable phases. For instance, Lee, Koh, Park and Ryu [48] analyzed the effects of high-temperature exposure on CAC-based UHPC, finding that C-A-(S)-H gel formation at 450 °C contributed to higher compressive strength, while C-A-S gel formation at 800 °C preserved residual strength through sintering. Kim, Son and Lee [49] highlighted challenges in the long-term durability of CAC due to high-temperature phase transitions, emphasizing the need for stabilization methods to improve the material's performance.

3D printing with calcium-based ACMs: The optimization of ACMs for 3D concrete printing has also been a major area of focus. In a recent study, Mohan, Rahul, De Schutter and Tittelboom [50] investigated CSA as an alternative binder for 3D concrete printing, finding that although CSA mixtures had higher pumping pressure due to greater plastic viscosity, partial substitution with limestone powder reduced this viscosity and pressure. Additionally, CSA-limestone

mixtures demonstrated improved buildability compared to Portland cement mixtures, which was linked to enhanced P-wave velocity growth at early ages.

Waste utilization in calcium-based ACMs: The use of waste materials in ACM formulations is expanding, offering a circular economy approach to construction. Further studies explored alternative aggregates and durability factors in CAC mixtures. Alizadeh, Rezaeian, Rasoolan and Tahmouresi [51] tested recycled ceramic aggregates (RCAs) in self-compacting concrete (SCC), discovering that RCA inclusion provided compressive strength gains at 800 °C but slightly reduced flowability and high-temperature strength in CAC-based mixes.

Self-healing calcium-based ACMs: Another notable advancement is the development of self-healing ACMs, which enhance the durability of concrete by improving its ability to repair cracks autonomously. Zokaei, Siad, Lachemi, Mahmoodi and Sahmaran [52] explored self-healing in CAC-engineered cementitious composites (ECCs), finding that CAC-ECCs with fly ash (FA) significantly improved early-age mechanical properties and recovery rates, even with phase conversion effects in non-FA mixes.

Application in engineered cementitious composites (ECCs): Engineered cementitious composites (ECCs) incorporating SCC have demonstrated sufficient strength and durability for various engineering applications, particularly when specific fly ash-to-SCC ratios are used. Zokaei, Siad, Lachemi, Mahmoodi, Ozcelikei and Şahmaran [53] demonstrated that while ECCs with SCC exhibit lower compressive and flexural strengths than traditional mixtures, they still meet most engineering requirements. Furthermore, recent innovations in chemical activation techniques have shown promise in enhancing the hydration characteristics and technical performance of both SCC and CSAB cements, supporting their use in sustainable construction practices [54–56].

Carbon capture and utilization: Finally, advancements in Carbon Capture and Utilization are playing a significant role in the development of low-carbon ACMs. By integrating CO₂ directly into calcium-based ACMs, these materials can form stable carbonates, helping to sequester carbon and reduce the environmental impact of construction. Shenbagam, Shaji, Eswita, Cepuritis and Chaunsali [57] investigated the carbonation behavior of a CSAB binder and found it to carbonate more rapidly than a PC binder. This rapid carbonation was accompanied by a reduction in compressive strength, which was attributed to an increase in pore volume caused by the carbonation of the ettringite-rich microstructure. This structural change facilitated further CO₂ ingress into the binder.

Alkali-activated materials (AAMs)

Alkali-Activated Materials (AAMs) are formed through the chemical reaction between alkaline activators—such as sodium hydroxide (NaOH), potassium hydroxide (KOH), or sodium silicate—and aluminosilicate-rich solid precursors like fly ash, slag, metakaolin, or natural pozzolans. The reaction process, often referred to as alkali activation or geopolymerization, results in a hardened binder with properties suitable for various construction applications. AAMs can be broadly categorized into three major subgroups based on their chemical structure, precursor composition, and the type of alkali activation process employed: Alkali-Activated Binders (AAB), Inorganic Polymers, and Geopolymers.

AABs are materials formed through the alkali activation of calcium-rich precursors, such as ground granulated blast-furnace slag (GGBS). The reaction predominantly involves the formation of calcium silicate hydrate (C-S-H) gels with additional contributions from aluminosilicate phases. Their main characteristics include high early strength (compressive strengths of up to 100 MPa), excellent durability and resistance to chemical attack, and low CO₂ emissions compared to traditional cement [58]. AABs offer high fire resistance, remaining stable up to 1000 °C, which can be further enhanced by fire-resistive coatings [59, 60]. They exhibit strong resistance to acid and sulfate attack, demonstrating superior chemical durability compared to ordinary Portland cement (OPC) [61, 62]. Strength, which varies based on precursor types and activators, can exceed 20 MPa early strength with proper curing [63, 64]. Their durability extends to aggressive chemical environments, resisting sulfate, carbonation, and chloride penetration, thus improving lifespan [65]. AABs have a significantly lower carbon footprint compared to Portland cement, with estimates indicating a 60–80% reduction in CO₂ emissions. AAB concrete also shows a 40–50% decrease in embodied carbon [66–68].

Inorganic polymers refer to a broader class of AAMs that exhibit polymer-like structures but are entirely composed of inorganic components. They are formed from aluminosilicate precursors activated under strongly alkaline conditions, leading to the development of a cross-linked network of silica and alumina tetrahedra. The structure of inorganic polymers involves a higher degree of silicate connectivity

compared to ordinary Portland cement. This can be represented by a higher ratio of (SiO₂+Al₂O₃) to (CaO+Na₂O), indicating a more complex silicate framework than what is typically observed in traditional ordinary Portland cement chemistry [69]. Inorganic polymers provide high resistance to fire and extreme temperatures, have good chemical and abrasion resistance, wide compositional flexibility, allowing customization for specific applications. They are mainly used in fire-resistant panels, high-temperature insulation, and waste encapsulation.

Geopolymers are a class of inorganic, amorphous, and cementitious materials formed by the alkaline activation of aluminosilicate-rich materials, such as fly ash, slag, metakaolin, and natural pozzolans [70]. Duxson, Fernández-Jiménez, Provis, Lukey, Palomo and van Deventer [66] describe the alkaline activation reaction in geopolymers as a series of steps: dissolution, condensation, polycondensation, and gel crystallization (Fig. 5). Initially, aluminosilicate materials dissolve in a strongly alkaline medium (pH > 14), breaking Si–O–Si and Al–O–Al bonds to release silicate ([Si(OH)₄][−]) and aluminate ([Al(OH)₄][−]) monomers, forming a colloidal phase. During condensation, unstable aluminosilicate species form, releasing water molecules and initiating gel formation. Alkali metal ions (e.g., Na⁺ or K⁺) play a crucial role in stabilizing these negatively charged gels by balancing their charges, allowing structural reorganization into stronger compounds. Polycondensation follows, where monomers bond through interactions between silicate OH groups and aluminate ions, forming stable gels. These gels can be amorphous (N-A-S-H) or crystalline (zeolites). The material subsequently hardens, developing mechanical strength and other characteristic properties of alkali-activated materials [71]. Geopolymer-based concrete can develop high early strength, with compressive strengths exceeding 40 MPa within the first 24 h of curing [66]. Furthermore, it exhibits excellent resistance to sulfate, acid, and high-temperature attacks, making it suitable for aggressive environments. It also exhibits low shrinkage and creep, leading to improved dimensional stability and reduced risk of cracking [72]. Geopolymers can achieve CO₂ emissions reductions of up to 40–80% compared to Portland cement, depending on the raw materials and production processes used [72, 73].

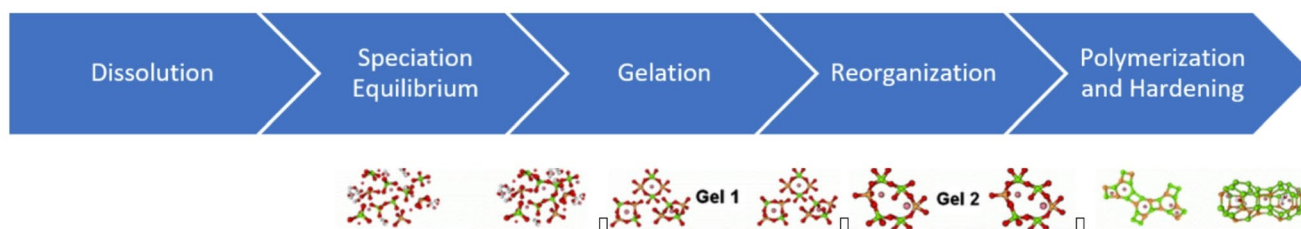


Fig. 5 Alkaline activation reaction in geopolymers [66]

While AAMs offer favorable early-age strength, chemical resistance, and environmental performance, concerns about their long-term durability—particularly carbonation and alkali-silica reaction (ASR)—remain active areas of research. Some studies have shown that AAMs, especially those based on low-calcium precursors, may be more susceptible to carbonation-induced pH reduction, potentially compromising steel reinforcement passivation in reinforced concrete applications. In addition, the high alkali content in certain AAM formulations can elevate the risk of ASR when reactive aggregates are used, unless carefully mitigated through precursor selection or the use of supplementary additives. These factors introduce uncertainty in service life prediction models, which must account for variable performance under different environmental conditions and mix designs. Ongoing research is needed to develop reliable modeling tools and long-term data to better predict the aging behavior of AAM-based concretes and to support their broader implementation in structural applications.

Advancement in AAMs

AAMs represent an exciting frontier in sustainable construction technology. While these materials offer substantial environmental advantages, their adoption has been limited by challenges in achieving consistent mechanical performance, ensuring long-term durability, and optimizing production processes. Recent advancements have focused on addressing these challenges to improve the practicality and scalability of AAMs for construction applications (Fig. 6).

Optimizing curing techniques and mix designs: Alkali-activated concrete (AAC) often requires elevated

temperatures or extended curing durations to attain desired mechanical properties. This necessity can complicate on-site construction processes, increase energy demands, and elevate overall costs. Moreover, long-term durability remains a significant concern, particularly in aggressive environments, such as acidic or sulfate-rich conditions. To address these limitations, researchers are exploring advanced mix designs, alternative curing techniques (e.g., ambient curing systems), and hybrid alkali activators that enhance reaction kinetics and durability under a broader range of conditions. These efforts aim to improve the performance, reliability, and cost-effectiveness of AAC.

Waste utilization in AAMs: The integration of industrial and agricultural waste as precursors in AAMs is a promising strategy for improving sustainability and reducing raw material costs. Over recent years, extensive research has explored the potential of various waste materials to serve as precursors or supplementary additives in AAMs. Examples include different construction and demolition waste [74], brick masonry waste [75], waste concrete powder [76], waste glass [77], marble powder [78], and municipal solid waste incineration bottom ash [79].

UHPC using AAMs: Recent studies have focused on developing cementless UHPC using alkali-activated materials, termed “ultra-high-performance geopolymers” (UHP-AAC) since its introduction in 2014 [80]. The initial UHP-AAC utilized ground granulated blast furnace slag (GGBFS) as precursors later followed by SF and FA. Other precursors, such as microsilica [81], metakaolin (MK) [82], rice husk ash (RHA) [83], and wheat straw ash (WSA) [84], have also gained attention recently. Different alkali activators, including NaOH, KOH, Na_2SiO_3 , Na_2CO_3 , K_2SiO_3 , and K_2CO_3 , have been utilized in UHP-AAC production. Aggregates such as quartz powder, sand, natural sand, and recycled materials like crushed glass, ceramic, and crumb rubber have been incorporated to reduce binder content and maximize packing density by optimizing particle size distribution and ingredient fractions [80, 85].

Nanomaterials application in AAMs: Another promising avenue in AAMs is the incorporation of nanomaterials. Several studies indicate that nanoparticles substantially enhance strength by facilitating the formation of dense gels and a strong surface morphology, thereby improving overall concrete performance. Moreover, nanomaterials can accelerate the hydration process, resulting in reduced setting times. Although the inclusion of nanomaterials may lead to decreased flowability, the trade-off is justified by the significant improvements in durability and mechanical properties [86, 87]. Recent innovations involving nanomaterials, such as nano-alumina, have demonstrated potential in enhancing both the fresh and hardened properties of alkali-activated

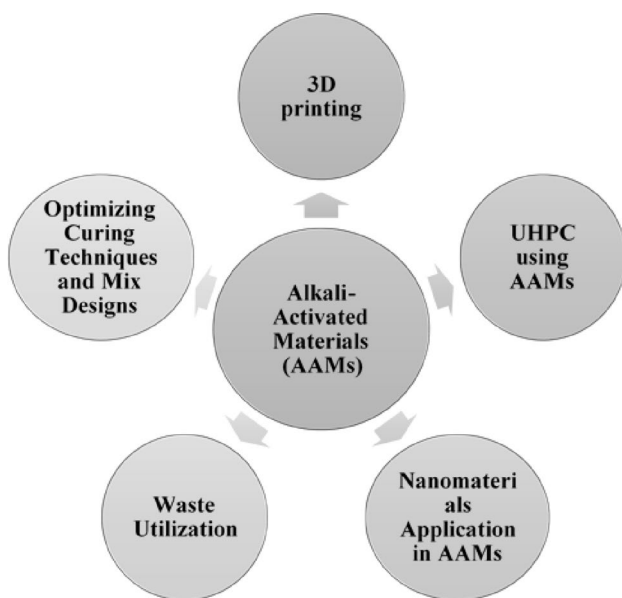


Fig. 6 Area of advancement for AAMs

self-compacting concrete (A-ASCC), which may result in more consistent and resilient performance [88].

3D printing with AAMs: Chen, Liu, Cao, Chen, Mechtcherine and Zhang [89] studied how precursor materials affect the rheology of alkali-activated materials (AAM) for 3D printing (3DPAAM). Using GGBFS, FA, and SF, they analyzed properties like static yield stress, thixotropy, and viscosity recovery. The results show that adding FA or SF increases static yield stress and influences structural build-up rates during flocculation and polycondensation, governed by inter-particle distance and precursor reactivity. Optimal dosages for improved 3D-printing rheology were identified as 25 wt% FA or 10 wt% SF, offering guidance for precursor selection and mixture design in 3DPAAM. In another study, Pavlin, Horvat, Cerc Korošec, Capuder, Korat and Ducman [90] investigated the use of AAM based on waste mineral wool for 3D printing applications. The study demonstrated the material's suitability by successfully producing 3D-printed samples. Key findings highlight the strong mechanical performance and thermal stability of these materials, making them suitable for high-temperature applications. This research offers a novel approach to utilizing waste mineral wool and co-binders in AAM development, contributing to advancements in construction materials and promoting sustainable recycling practices.

Magnesium-based ACMs

Magnesium-based alternative cementitious materials encompass Magnesium Phosphate Cements (MPCs), Magnesium Oxychloride Cement (MOC), and Blended Magnesium-Based Systems each offering unique properties and applications in construction and waste confinement.

MPCs typically form through an acid–base reaction between MgO and phosphate sources, resulting in the precipitation of crystalline phases like K-struvite, which contribute to rapid strength gain, with compressive strengths often exceeding 40 MPa within hours of setting, which is considerably faster than Portland cement [18, 91]. The high early strength and stable later strength development make MPCs suitable for rapid repairs and structural applications [92, 93]. The inclusion of borates and other chemical modifiers allows for precise control of setting times, ensuring broad applicability in diverse construction scenarios. These cements can be synthesized from sources such as low-grade MgO, which is cost-effective and sustainable without compromising mechanical strength or mineralogical stability. MPCs, offer excellent fire resistance, acting as fire-retardant coatings that protect substrates by maintaining temperatures below 550 °C for two hours. Their effectiveness is due to stable crystalline phases and heat-absorbing free water [94–96]. MPCs exhibit good chemical erosion

resistance, particularly against sulfate and acidic environments. They show superior resistance in solutions with magnesium sulfate and sodium sulfate compared to alkaline solutions where resistance decreases [97–99]. Additives such as metakaolin can improve MPC's pore structure and water resistance, enhancing durability against corrosion [100]. The production of MPCs can reduce carbon emissions significantly. Utilizing salt lake magnesium residues to produce MgO precursors for MPCs can cut CO₂ emissions by over 60% compared to traditional Portland cement [101, 102]. Some MPC types are carbon-negative, actively sequestering CO₂ during carbonation curing [103]. MPCs may face challenges with fast initial hydration and poor water resistance, which can hinder long-term durability in moist environments [104]. Adjustments with retarders and polymer additives are often needed to control setting times and improve cohesiveness [18].

MOC, also known as Sorel cement, results from the reaction between MgO and MgCl₂ in water, producing several hydration phases such as phase 3 (3 Mg(OH)₂·MgCl₂·8H₂O) and phase 5 (5 Mg(OH)₂·MgCl₂·8H₂O), which form needles or sheets imparting high strength and fire resistance properties [105]. MOC exhibits excellent fire resistance due to abundant crystalline water and MgO's high melting point (~2800 °C), which act as heat insulators. It outperforms Portland cement in durability under high temperatures and is commonly used in fire protection boards [106–108]. MOC shows excellent resistance to salt attack, including corrosive brine and aging solutions, making it suitable for harsh environments. It is more resistant to chloride and sulfate penetration compared to Portland cement [109]. MOC develops high compressive strength rapidly, reaching values above 90 MPa within 28 days when prepared with optimized calcination and MgCl₂ concentration [110]. The strength is closely linked to the proportion of Phase 5 crystals formed during hydration [111]. Although MOC has good mechanical properties and chemical resistance, its poor water resistance limits its application in wet conditions [112, 113]. Water exposure can degrade hydration phases, resulting in strength loss and material deterioration [114, 115]. MOC has also undergone significant modifications, including the incorporation of additives to enhance water resistance and reduce brittleness. High-performance MgO-based panels, made using fibers or fillers, are now being developed for construction and insulation applications, showcasing the versatility of these materials in addressing modern construction needs [116]. These phases contribute to the cement's superior mechanical properties in comparison to Portland cement but face challenges like poor long-term water resistance due to leaching of MgCl₂ [117, 118]. MOC production reduces energy consumption and CO₂ emissions compared to Portland cement since calcination temperatures are

lower (~55% of OPC). Furthermore, MOC can absorb CO₂ through carbonation, offsetting emissions [110, 119, 120].

Blended Magnesium-Based Systems, combining MgO with silica sources such as metakaolin, rice husk ash, or fly ash facilitates pozzolanic reactions, leading to the formation of magnesium-silicate-hydrate (M-S-H) gels. These gels enhance mechanical strength and durability. Additionally, blending MgO with calcium sulfoaluminate-belite (CSAB) cements improves resistance to sulfate attack and carbonation, broadening the usability of magnesium-based ACMs in harsh environments [121, 122]. Blended magnesia-based cements benefit from the inherent fire resistance of magnesia, contributing to better thermal stability and insulation in composite materials [111]. The integration of magnesia can enhance resistance to sulfates and chloride erosion depending on the blend composition and curing conditions [123]. Blending MgO with Portland cement accelerates early strength development and enhances microstructural formation, especially when supplementary cementitious materials such as fly ash or slag are added [124, 125]. These systems generally show improved durability with lower permeability and enhanced resistance to chemical attacks, though the long-term performance depends on specific blend ratios and additives used [125]. Magnesium-based residues, such as byproducts from Salt Lake lithium extraction or other industrial processes, can be processed and used as raw materials in cement production. Using these residues not only diverts waste from disposal but also avoids the carbon emissions associated with producing virgin raw materials. Studies show that preparing MgO from salt lake magnesium

residues can reduce CO₂ emissions by over 60% compared to traditional Portland cement production [102]. Incorporating these residues into blended cement formulations helps further decrease the cement's overall embodied carbon footprint [126].

Advancement in magnesium-based ACMs

Below are the key advancements in this field (Fig. 7):

Reactive MgO production: The controlled calcination process for reactive MgO production enhances its hydration reactivity and carbon sequestration potential. This dual-purpose material supports both structural performance and environmental sustainability, making it a pivotal development in magnesium-based ACMs [127].

Activation of magnesium phases: Another notable innovation involves the activation of hydro-magnesite and magnesite phases. These mineral phases contribute to improved early-age strength development and long-term stability in cementitious systems.

Utilization of industrial by-products: Industrial waste streams, such as brine and seawater, are now being repurposed as sources for magnesium compounds. This innovative approach reduces waste and promotes circular economy practices, enhancing the overall sustainability of magnesium-based ACMs [128].

Applications in specialty concrete and 3D printing: Magnesium-based ACMs are increasingly used in specialty concrete and 3D printing applications. Their rapid setting characteristics and adjustable rheology make them suitable for complex and innovative construction techniques, such as 3D concrete printing [129, 130].

Challenges addressed: Water resistance issues in MOC have been partially mitigated through hybrid additives and carbonation curing [111, 131]. Fast setting and short workability times in MPCs are managed by retarders and admixtures to extend application ability [93, 132]. Efforts continue to optimize blend compatibility, hydration mechanisms, and long-term durability for broad industrial adoption [133].

Emerging applications and industrialization: Magnesium-based cements are increasingly integrated into precast building materials, fireproof panels, rapid repair mortars, and biocompatible bone cements [134]. Use of renewable or waste-derived magnesium sources and development of patented manufacturing technologies support cost-effective and sustainable industrial scaling [102].

Activated glassy cements (AGC)

Activated Glassy Cements (AGC) are a class of alternative cementitious materials that are produced by the activation of calcium-rich glassy materials, such as blast furnace

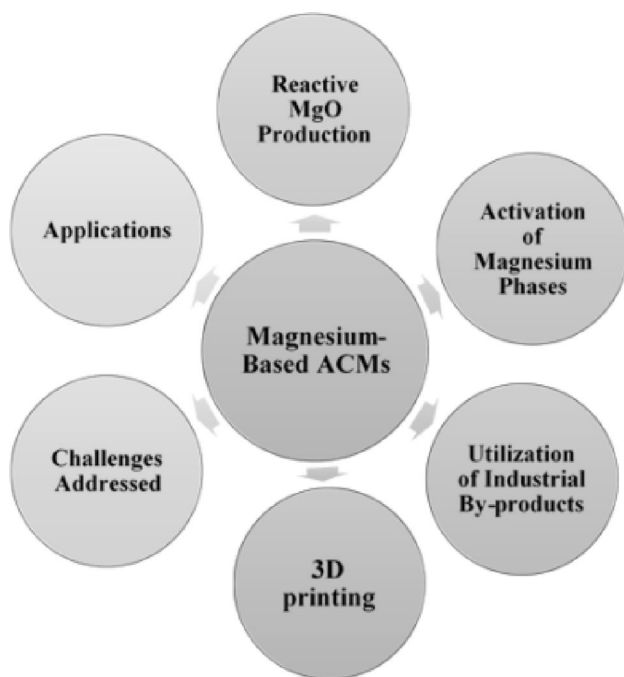


Fig. 7 Area of advancement for magnesium-based ACMs

slag, steel slag, or waste glass [135]. The activation process breaks down the glassy structure and allows the formation of calcium-silicate-hydrate (C-S-H) and other hydration products, similar to those found in Portland cement-based concrete. AGC-based concrete can develop high early and long-term compressive strengths, exhibits enhanced resistance to chemical attack, and reduced shrinkage and creep [135–137]. They resist high temperatures up to 1100 °C, significantly higher than conventional Portland cement which fails above 200 °C [138]. AGCs can reduce CO₂ emissions by 57–83% compared to Portland cement, achieving 3.8–4.9 times lower emissions than recycling waste glass alone [139].

Advancement in AGCs

Below are the major advancements categorized into distinct areas (Fig. 8):

Hybrid mixtures: A major drawback of using waste glass in concrete production is its tendency to weaken and crack the concrete due to expansion caused by the alkali-silica reaction (ASR). Park and Lee [140] examined how waste glass color (amber and emerald-green) and industrial by-products (ground granulated blast-furnace slag, fly ash) influence ASR. Results showed that the optimum size for ASR expansion was 2.5–1.2 mm, with smaller particles (<1.2 mm) exhibiting reduced expansion. Incorporating industrial by-products helped mitigate ASR-induced expansion and strength loss. Additionally, finely ground glass (<0.075 mm) was found to be a viable pozzolanic material, enhancing concrete performance.

Particle size optimization: Research on particle size optimization has shown that advanced milling techniques can produce ultra-fine glass powders with a higher surface area. These powders exhibit increased reactivity, contributing significantly to strength development and enhancing the overall performance of AGCs [141].

Carbonation curing for sustainability: Incorporating carbonation curing into AGC systems enables the capture and storage of CO₂ during the curing process. This method converts calcium hydroxide and alkali phases into stable carbonates, thereby improving both sustainability and material properties. This advancement aligns with global goals of reducing the carbon footprint of construction materials.

Waste utilization: Park, Lee and Kim [142] analyzed waste glass using SEM and EDX to examine the elemental composition. The analysis revealed that silica oxide (SiO₂) had the highest concentration in the chemical composition test (71.30–73.04%), followed by Na₂O and K₂O. Minor variations in chemical composition were observed among waste glass samples of different colors. Additionally, they reported that the typical grain shape across all waste glass colors was angular. Tuam, Shitote and Oyawa [143] examined the use of waste glass as a replacement for fine aggregate. They concluded that incorporating up to 30% recycled glass into self-compacting mortar mixes is feasible, providing comparable strength performance, improved sorptivity, and no long-term adverse effects from ASR. In another study, Afshinnia and Rangaraju [144] investigated the effects of ground glass powder as a replacement for cement or coarse aggregate on the fresh and mechanical properties of Portland cement concrete. Using glass powder as a cement replacement consistently reduced compressive strength, regardless of the aggregate type. However, when glass powder replaced coarse aggregate, the compressive strength varied: it increased with crushed glass aggregate but decreased with natural mineral aggregate.

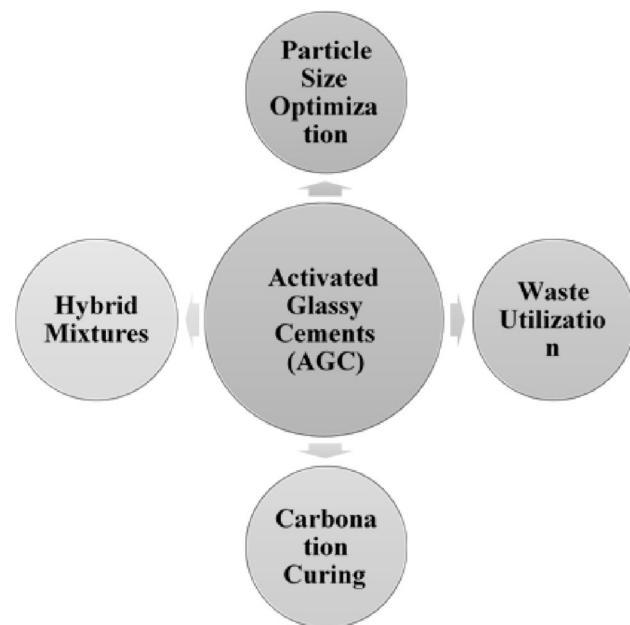


Fig. 8 Area of advancement for activated glassy cements (AGC)

CO₂-cured cement

CO₂-cured cement, also known as CO₂-sequestering cement or CO₂-mineralized cement, is an emerging type of alternative cementitious material that utilizes the injection of carbon dioxide (CO₂) during the curing process to enhance the properties and environmental performance of the final concrete product. The introduction of CO₂ during curing triggers a series of chemical reactions, leading to the formation of calcium carbonate (CaCO₃) and other stable hydration products [145, 146]. CO₂ cured concrete exhibits better fire resistance at elevated temperatures (e.g., 800 °C) compared to conventionally cured concrete [147, 148]. CO₂ curing improves corrosion resistance and durability against acid and sulfate attacks, increasing electrical resistivity

and compactness [149–151]. CO₂ curing accelerates early strength development, with increases up to 47.2% at early ages; concrete can reach about 76% of the target strength within 2 h [152–155]. CO₂-cured cement is its ability to sequester and store atmospheric CO₂ within the concrete matrix, effectively reducing the overall carbon footprint of the construction industry [156, 157] by approximately 30% in cement production; however, comprehensive life-cycle assessments show mixed results, with 56–68 out of 99 datasets indicating potential negative net CO₂ benefits primarily due to increased OPC use and energy penalties for CO₂ capture [158]. The carbonation process can potentially enhance the recyclability of CO₂-cured cement-based concrete, as the formed calcium carbonate can be more easily recovered and reused. The use of CO₂ as a raw material in the production of CO₂-cured cement can help divert industrial CO₂ emissions from various sources, contributing to a more sustainable and circular economy [156, 157].

Advancement in CO₂-cured cement

Below are the major advancements categorized into distinct areas (Fig. 9):

Carbonation curing for strength and durability: This innovative curing method enhances mechanical properties such as compressive strength while also increasing the material's resistance to environmental factors. Furthermore, carbonation curing offers a dual benefit by sequestering CO₂ within the concrete matrix, thereby contributing to global efforts to mitigate climate change [159].

Effectiveness of ready-mix CO₂-cured concrete: A review of CO₂-cured ready-mix concrete (CRC) technologies highlights their effectiveness in balancing workability,

mechanical properties, and durability. CRC has been found to exhibit superior performance metrics while significantly reducing the carbon footprint of concrete production. Life Cycle Assessment (LCA) studies underscore its potential to contribute to sustainable construction practices by integrating CO₂ utilization with enhanced material performance [160].

Environmental and economic considerations: While CO₂ curing offers clear advantages, comprehensive life cycle assessments have shown that some CO₂-cured concretes may result in higher overall CO₂ emissions if increased amounts of cementitious materials are required to compensate for strength reductions caused by curing. Ensuring that CO₂ curing consistently enhances compressive strength and reducing the energy consumption of the curing process are critical to maximizing carbon benefits [158]. The process also permanently stores CO₂ as calcium carbonate within the concrete matrix, turning concrete components into long-term carbon sinks and thereby contributing to carbon sequestration goals.

Table 1 provides a summary of advantages, limitations, and carbon emissions reduction of different ACMs.

Alternative clinkers

Alternative clinkers are materials used in the production of cement that serve as substitutes for traditional Portland cement clinker. These clinkers often incorporate industrial by-products, waste materials, or alternative raw materials to reduce the environmental impact of cement production. Some the most well-knowns alternative clinkers with respect to global reductions in concrete-related CO₂ emissions are Reactive Belite-rich Portland cement (RBPC) clinkers, Belite-Ye'elinite-Ferrite (BYF) clinkers, Carbonatable Calcium Silicate clinkers (CCSC), and Magnesium Oxides derived from Magnesium Silicates (MOMS) [29]. These alternative clinkers significantly cut concrete-related CO₂ emissions globally by reducing reliance on the highly carbon-intensive traditional clinker production process [162]. Such materials introduce novel chemistries that can lower the carbon footprint of cement without compromising its binding properties, thus contributing to the sustainability of the construction industry.

Traditional clinker production, however, faces significant environmental and operational challenges. It accounts for approximately 5%–8% of total anthropogenic greenhouse gas emissions, primarily due to the calcination process where limestone (calcium carbonate) decomposes into calcium oxide and emits large quantities of CO₂, and the high energy requirements of heating kilns to nearly 1450 °C. The process exhibits high fuel consumption (60–30 kg per ton of

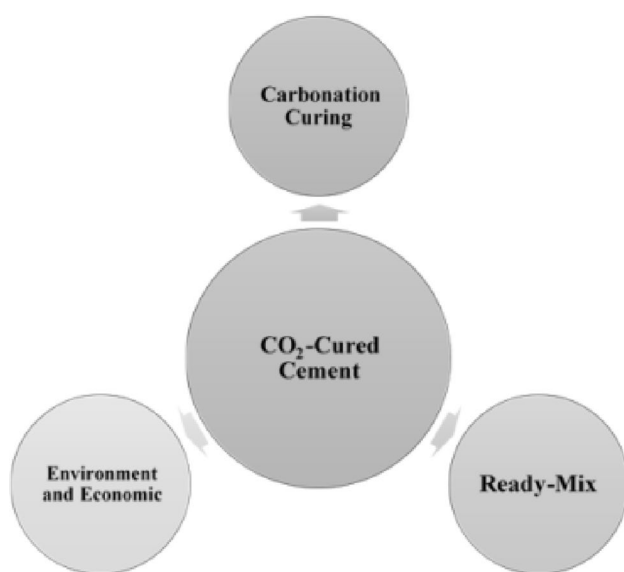


Fig. 9 Area of advancement for CO₂-cured cement

Table 1 Summary of ACMs advantages, limitations, and carbon emission reduction

Alternative cementitious materials	Advantages	Limitations	Carbon footprint reduction
Calcium-Based ACMs			
Calcium Aluminate Cement (CAC) [161]	Rapid early strength Resistance to chemical attack Lower firing temperatures More easily recyclable	High cost Susceptibility to conversion at elevated temperatures	5–20% lower CO ₂ emissions
Calcium Sulfoaluminate Cement (CSA) [31]	Rapid early strength Improved dimensional stability Resistance to chemical attack	Limited long-term strength Potential volume instability	30–40% lower CO ₂ emissions 20–30% lower embodied carbon
Super Sulfated Cements (SSC) [29]	Resistance to chemical attack Low heat of hydration	Slower strength development Limited availability of raw materials	49–95% lower CO ₂ emissions
Calcium Sulfoaluminate Belite (CSAB) Cements [29]	Rapid early strength Improved dimensional stability Resistance to chemical attack	Limited long-term strength development	30–50% lower CO ₂ emissions
Alkali-Activated Binders (AAB) [66, 67]	High early and long-term strength Resistance to chemical attack Thermal and fire resistance No limestone or high-temperature kiln	Requirement for specialized alkaline activators Potential for efflorescence	60–80% lower CO ₂ emissions 40–50% lower embodied carbon
Geopolymers [70]	High early and long-term strength Resistance to chemical attack Thermal and fire resistance Absence of limestone calcination and the lower energy requirements	Requirement for specialized alkaline activators Potential for efflorescence	40–80% lower carbon emissions
Magnesium-Based ACMs			
Magnesium Oxychloride and Oxysulfate Cements (Sorel Cements)	Rapid early strength gains Excellent fire resistance Enhanced Resistance to chemical attack	Limited availability of suitable raw materials Diminished performance in high humidity or wet environments	50–60% lower CO ₂ emissions
Magnesium Phosphate Cements (MPC)	Lower calcination temperatures Carbon sequestration potential		
Magnesium Hydroxycarbonate Cements			
Magnesium Oxide Cements (MOC) [117, 118]			
Activated Glassy Cements (AGC) [136, 137]	Improved durability Reduced environmental impact Lower temperatures and less limestone	Slower strength development compared to traditional Portland cement	57–83% lower CO ₂ emissions, carbon capture and sequestration
CO ₂ -cured Cements [156, 157]	Potential for carbon capture and storage Improved durability	Specialized curing equipment and process	30% lower CO ₂ emissions

cement) and electrical energy demand (~ 110 kWh per ton). Besides the environmental burden, the traditional process is capital- and energy-intensive, with heavy regulatory and logistical constraints limiting rapid transitions to greener technologies. Furthermore, since cement is generally a low-value, bulk commodity often sold and consumed domestically, opportunities for large-scale trade-based decarbonization are minimal, which complicates diffusion of best practices across regions.

The urgency to develop and adopt alternative clinker technologies is driven by the need to meet global climate targets, including net-zero greenhouse gas emissions by 2050, while accommodating increasing cement demand in developing economies. Cement production emissions currently amount to about 6% of global greenhouse gases and are projected to remain high or even double in some regions by the end of the century without substantial innovation. Despite some advancements in reducing clinker content via

blends, substantial emission cuts require breakthroughs in alternative clinker types such as those mentioned, alongside process electrification and carbon capture techniques. Bridging this innovation gap is critical since existing infrastructure lock-ins and the scale of global demand present formidable challenges. Policy incentives, financial investments, and coordinated industry efforts will be paramount to accelerate large-scale deployment and ensure sustainable cement and concrete production worldwide [162].

Reactive Belite-rich Portland cement (RBPC) clinkers

Reactive Belite-rich Portland Cement (RBPC) clinker is a specialized variant of Portland cement distinguished by its high belite (dicalcium silicate, C_2S) content. This enhanced belite concentration is achieved through the precise optimization of clinker composition and pyro-processing conditions, involving a partial substitution of the more energy-intensive alite (tricalcium silicate, C_3S) with belite. The result is a cement that not only enhances reactivity but also improves sustainability by significantly lowering the carbon footprint during production while maintaining desirable performance characteristics [29].

RBPC-based concrete exhibits enhanced resistance to chemical attacks, including sulfate exposure and alkali-silica reactions. Its belite-rich composition contributes to reduced shrinkage and improved dimensional stability. RBPC clinkers reduce CO_2 emissions primarily by lowering the limestone content and altering clinker phases to belite, which requires less calcination energy. Studies indicate that RBPC can reduce process-related CO_2 emissions by around 10% compared to ordinary Portland cement (OPC). However, this reduction is relatively modest compared to other alternative clinker technologies, and achieving greater environmental goals requires additional innovations [29, 137, 163–165].

Belite-Ye'elimite-ferrite (BYF) binder technology

The Belite-Ye'elimite-Ferrite (BYF) binder technology is based on clinkers comprising three essential phases: Belite (Dicalcium Silicate, C_2S), Ye'elimite (Calcium Sulfoaluminate, CSA, $C_4A_3\$$), and Ferrite (Calcium Alumino-Ferrite, Brownmillerite, C_4AF) [29]. The BYF approach can be viewed as an intermediate technology between conventional OPC and the established CSA cement technologies. One of the primary advantages of BYF cements is that they can be produced in standard Portland cement plants, which reduces capital investment costs. The inclusion of ye'elimite in BYF clinkers facilitates rapid hydration and early strength gain in BYF-based concrete. The belite-rich composition of these clinkers allows for moderate long-term strength development

compared to traditional Portland cement. Moreover, BYF-based concrete displays enhanced resistance to sulfate and chloride attacks, thanks to its unique phase composition. Additionally, the belite-rich nature of BYF clinkers contributes to reduced shrinkage and creep in the resulting concrete [28, 29]. BYF clinkers present a more substantial reduction in carbon emissions, balancing clinker chemistry to reduce limestone usage and firing temperature. Studies report CO_2 emission reductions ranging from approximately 28%–42% versus OPC. The BYF system also shows promise in stabilizing heavy metals, improving environmental performance beyond CO_2 reduction [166, 167].

Carbonatable calcium silicate clinkers (CCSC)

CCSC is an alternative cement primarily composed of reactive calcium silicate phases, such as dicalcium silicate (C_2S) and tricalcium silicate (C_3S). What sets CCSC apart is its ability to carbonate, where calcium silicate phases react with atmospheric carbon dioxide (CO_2), forming calcium carbonate ($CaCO_3$) and calcium-silicate-hydrate (C-S-H) gel. This carbonation process enhances long-term strength and durability by mitigating shrinkage and creep, resulting in improved dimensional stability of CCSC-based concrete [137]. The carbonation process in CCSC not only strengthens the concrete but also sequesters atmospheric CO_2 , significantly reducing the material's carbon footprint. The production of CCSC requires lower calcination temperatures compared to traditional Portland cement, leading to lower energy consumption and reduced CO_2 emissions. Additionally, CCSC can be produced using industrial by-products such as fly ash, slag, and limestone, further enhancing its sustainability [29].

Recent developments in CCSC technology focus on accelerating and controlling the carbonation process in industrial applications without excessive energy use. Studies have shown that binders made from simple calcium silicates, such as wollastonite ($CaSiO_3$), carbonate rapidly in CO_2 -rich environments [168]. This makes it feasible to manufacture precast CCSC-based concrete in non-pressurized carbon dioxide curing chambers, with controlled ventilation, temperature, and humidity. While this method requires some capital investment, the process is more energy-efficient as no heating is necessary due to the exothermic nature of carbonation. However, water vapor removal requires refrigeration, adding some operational costs.

Advantages of CCSC Technology can be listed as low CO_2 emissions, high CO_2 absorption, rapid strength gain, and low water consumption [169]. CCSC have been demonstrated to achieve substantial reductions in CO_2 emissions, typically in the range of 30–50% compared to ordinary Portland cement [170–172]. This reduction arises from lower

clinker content, reduced calcination temperatures, and CO₂ uptake via carbonation curing. The ability to sequester CO₂ during curing further amplifies emissions savings. CCSC-based cements thus represent a promising strategy for decarbonizing the cement industry while meeting performance requirements [170]. CCSC is mainly suited for the production of precast concrete articles that are not too large in cross-section to ensure complete curing. Its low pH (~9) makes it unsuitable for protecting mild steel from corrosion in humid environments, especially in the presence of chloride or sulfate ions [29].

Magnesium oxides derived from magnesium silicates (MOMS)

MgO is primarily produced by calcining natural magnesite (MgCO₃), a process that releases substantial amounts of CO₂—about 52% of magnesite's mass is released as RM-CO₂, in addition to emissions from the energy-intensive calcination process. In concrete applications, MgO hydrates to form magnesium hydroxide (brucite, Mg(OH)₂). However, brucite alone forms weak pastes, as it lacks strong bonding properties. To improve performance, MgO can be combined with magnesium salts, such as magnesium chloride or magnesium sulfate, to produce Sorel cements. These Sorel-based materials have been used for over a century in the manufacture of lightweight building panels with superior strength and fire resistance compared to gypsum-based products. However, they lose strength rapidly in high humidity or when exposed to water, making them unsuitable for moist environments [117]. Alternatively, combining MgO with phosphate salts results in highly durable, water-resistant pastes. However, phosphate is a scarce resource primarily reserved for agriculture, making these binders too costly for large-scale construction, restricting them to niche, high-performance uses.

A noteworthy advancement in the realm of MgO-based cements is the emergence of magnesium hydroxy-carbonate cements, pioneered by Vlasopoulos and Cheeseman [173]. These cements offer improved water resistance and the ability to sequester CO₂ during hydration, theoretically offsetting the CO₂ emissions from production. However, achieving “low-CO₂” status is only possible if the MgO is derived from magnesium sources with no inherent RM-CO₂, such as magnesium silicate rocks (e.g., serpentines and olivines), which are abundant globally but not as easily accessible as limestone [174]. Despite these sourcing considerations, materials formulated with magnesium oxides derived from magnesium silicates hold significant promise for substantial CO₂ emissions reductions when compared to ordinary Portland cement. Estimates suggest these reductions could range from 50 to 70%. This impressive decrease stems from

several key factors: lower calcination temperatures required for processing, reduced CO₂ emissions from limestone decarbonation (as limestone isn't the primary magnesium source), and critically, the CO₂ uptake during carbonation curing [175, 176]. The integration of carbonation processes into the curing phase further enhances carbon capture, ultimately leading to a significantly lower carbon footprint for these advanced cements.

Aside from hydraulic binders, MgO can also harden via direct carbonation, capturing significant amounts of CO₂ under modest pressure, similar to the Carbonatable Calcium Silicate Cement (CCSC) approach. This technology holds potential for precast products where controlled curing environments are feasible [177].

The main challenge for MOMS (Magnesium Oxides from Magnesium Silicates) technology is developing an energy-efficient method for producing MgO from magnesium silicate rocks at an industrial scale. If research funding addresses these challenges, MOMS binders could theoretically achieve negative net CO₂ emissions (i.e., net CO₂ capture), positioning them as a promising low-carbon alternative for the construction industry [178]. This potential makes MOMS technology a significant candidate for future advancements in sustainable concrete applications.

Limestone calcined clay cement (LC3)

Limestone Calcined Clay Cement (LC3) represents a cutting-edge advancement in sustainable cement technology, offering an environmentally friendly alternative to traditional Portland cement. LC3 is characterized by its ternary composition, where up to 50% of the clinker (the primary energy-intensive component of cement) is replaced with limestone and calcined clay. Typically, clinker replacement with limestone is limited to around 15%, while calcined clay replacement can reach up to 30%. The production process involves either co-grinding or blending these components to create a homogeneous mixture, balancing performance with sustainability [179].

The primary reactions in LC3 arise from interactions between calcined clay, limestone, and clinker hydration. Two key reaction pathways are:

Pozzolanic reaction of calcined clay with calcium hydroxide: Calcined clay, mainly metakaolin obtained by heating kaolinite-rich clay above 700 °C, provides reactive alumina. This alumina reacts with calcium hydroxide (Ca(OH)₂), a product of clinker hydration, forming calcium aluminate hydrates. These hydrates enhance cement strength and reduce porosity.

Carbonation reaction of limestone with alumina: Calcium carbonate from the finely ground limestone reacts with the reactive alumina released from calcined clay to form

carboaluminate hydrates. These hydrates fill pore spaces, contributing to improved durability and long-term stability.

Together with the conventional clinker hydration that forms calcium silicate hydrates (CSH), these reactions create a synergistic effect. The combined formation of CSH, calcium aluminate hydrates, and carboaluminate phases leads to LC3's superior mechanical performance and durability compared to ordinary Portland cement (OPC).

The selection and treatment of raw materials critically influence LC3's reactivity and performance. Clays rich in kaolinite show high pozzolanic activity compared to clays containing illite or montmorillonite, due to their mineralogical structure. Calcination at temperatures above 700 °C removes chemically bound water from these clays and

disrupts their crystalline structure, increasing their reactivity in the high-pH cementitious environment [180].

LC3's lower clinker content results directly in reduced CO₂ emissions during production. Additionally, the formation of calcium aluminate and carboaluminate hydrates enhances mechanical properties and improves durability, making LC3 a promising, sustainable cement alternative. In comparison to OPC, LC3 is capable of reducing carbon emissions by 30% and conserving more than one-third of energy usage [181]. Table 2 provides a summary of information related to different alternative clinkers.

Supplementary cementitious materials (SCMs)

Supplementary Cementitious Materials (SCMs) have gained significant attention in recent years as a viable solution for developing low-carbon concrete. SCMs are materials that can partially replace traditional Portland cement in concrete mixes, thereby reducing the overall carbon footprint of the final product. By incorporating these supplementary materials, carbon emissions associated with cement manufacturing can be significantly diminished, while also enhancing the durability and performance of concrete [13].

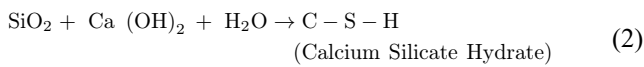
The use of industrial and agricultural wastes as SCMs can promote the environmental sustainability of the concrete industry by minimizing the negative impact of the explosive usage of raw materials. These waste materials, such as silica fume, fly ash, slag cement, rice husk ash, palm oil fuel ash (POFA), bagasse ash (BA), wood waste ash, bamboo leaf ash (BLA), corn cob ash (CCA), and metakaolin (calcined clay) can be used as alternatives to Portland cement by leveraging their engineering, physical, and chemical properties.

Silica fume

Silica fume, also known as microsilica, is a by-product of the silicon metal or ferrosilicon alloy industries. Composed of extremely fine, spherical particles of amorphous silicon dioxide (SiO₂), it is a highly reactive pozzolanic material. With particles measuring between 0.1 and 0.5 microns and an exceptionally high surface area, silica fume's reactivity in cementitious systems is significantly higher than that of many other SCMs. This fine particle size enables it to enhance various properties of concrete, including compressive strength and durability. The increased strength is primarily due to the pozzolanic reaction, where silica fume reacts with calcium hydroxide (Ca (OH)₂), a by-product of cement hydration, forming additional calcium silicate hydrate (C-S-H), which strengthens the concrete matrix. This reaction can be represented as:

Table 2 Summary of alternative clinkers advantages, limitations, and carbon emission reduction

Alternative clinkers	Advantages	Limitations	Carbon footprint
Reactive Belite-rich Portland Cement (RBPC) Clinkers [163]	Reduced carbon emissions during production Improved durability and resistance to chemical attack	Slower strength development compared to traditional Portland cement	Lower lime saturation factor Higher belite (C ₂ S) content Reduced energy consumption 10% lower CO ₂ emissions [29, 165]
Belite-Ye'elime-Ferrite (BYF) Clinkers [163]	Rapid early strength development Reduced CO ₂ emissions during production	Limited long-term strength development	Lower calcination temperatures Less limestone CO ₂ emission reductions ranging from approximately 28%–42% [166, 167]
Carbo-natable Calcium Silicate Clinkers (CCSC) [169]	Potential for carbon capture and storage Improved durability	Limited availability of raw materials	Lower calcination temperatures Incorporation of Industrial By-products 30–50% lower CO ₂ emissions [170–172]
Magnesium Oxides Derived from Magnesium Silicates (MOMS) [182]	Improved durability Higher water resistance	Energy-Intensive Processing Slow Reactivity Cost and Availability	Lower calcination temperatures CO ₂ Sequestration Potential Incorporation of Industrial By-products 50–70% lower CO ₂ emissions [175, 176]
Limestone Calcined Clay Cement (LC3) [179]	Significant CO ₂ Emission Reduction Enhanced Durability Lower Cost of Production	Availability of High-Quality Clay Early Strength Development Processing and Quality Control Lack of Standardization	30% lower CO ₂ emissions [181]



This reaction densifies the microstructure of the cement paste, reducing porosity and filling voids within the concrete. As a result, the concrete achieves higher compressive strength, as well as improved resistance to chemical attacks such as those from chlorides and sulfates [183, 184].

In terms of environmental benefits, the partial replacement of Portland cement with silica fume significantly reduces the embodied carbon of concrete. The production of silica fume has a lower carbon footprint compared to traditional cement manufacturing, contributing to the reduction of greenhouse gas emissions. Additionally, its use in concrete can reduce the overall energy consumption and waste generation associated with the construction industry. When combined with other SCMs like fly ash and ground granulated blast-furnace slag (GGBS), silica fume can further reduce the environmental impact of concrete, making it a valuable tool in the pursuit of sustainable construction practices [185, 186]. Typically, silica fume is used in proportions ranging from 5 to 15% of the total cementitious material in concrete. While 5% is generally the minimum to observe significant improvements in strength and durability, the upper limit of 10–15% ensures these benefits without negatively impacting workability. In some specialized applications, higher percentages may be used, though this requires careful adjustments in mix design to prevent issues like increased water demand or brittleness [187].

Fly ash

Fly ash is a fine, powdery by-product generated from the combustion of coal in power plants. This material is classified as a pozzolanic material, meaning it reacts with calcium hydroxide ($\text{Ca}(\text{OH})_2$) in the presence of water to form additional calcium silicate hydrate (C-S-H), which contributes to the strength and durability of concrete. Fly ash is primarily composed of silicon dioxide (SiO_2), aluminum oxide (Al_2O_3), and iron oxide (Fe_2O_3), with its specific chemical and physical properties varying based on the type of coal burned and combustion conditions. These variations influence its performance in concrete applications, leading to the classification of fly ash into two categories: Class F and Class C [187, 188]. Class F fly ash is derived from the combustion of harder, older anthracite and bituminous coals. It has lower calcium oxide (CaO) content, typically less than 10%, making it a low-lime pozzolan. Due to its lower calcium content, Class F fly ash is primarily a pozzolanic material and relies heavily on the presence of calcium hydroxide for strength development. Class C fly ash, derived from the combustion of younger, sub-bituminous, or lignite coals,

has higher calcium oxide content, typically above 20%, giving it both pozzolanic and cementitious properties. Unlike Class F, Class C fly ash can form C-S-H on its own, without the need for additional calcium hydroxide. This makes it suitable for applications where early strength gain is desired [183].

Slag cement

Slag cement, commonly referred to as ground granulated blast-furnace slag (GGBFS), is a highly sustainable cementitious material created by grinding granulated blast-furnace slag, a by-product of the iron and steel industry. Its composition predominantly consists of calcium, silicon, and aluminum oxides, which provide both pozzolanic and latent hydraulic properties, making it a valuable ingredient in concrete mixes [187]. The chemical composition and fineness of slag cement can vary depending on the source and production process, but these characteristics significantly contribute to its performance in concrete applications [184].

Agricultural ashes

Rice husk ash

Rice Husk Ash (RHA) is a pozzolanic material obtained by the controlled combustion of rice husks, a by-product of the rice milling industry. RHA is composed primarily of amorphous silica, which gives it the ability to react with calcium hydroxide to form additional cementitious compounds. The chemical and physical properties of RHA can vary depending on the combustion conditions and the source of the rice husks.

The incorporation of RHA can improve the workability of fresh concrete due to its fine particle size and spherical shape, which help reduce water demand. RHA can contribute to the long-term strength development of concrete through its pozzolanic reaction with calcium hydroxide, which densifies the cement paste matrix [189, 190]. The use of RHA in concrete can improve its resistance to various forms of deterioration, such as sulfate attack, alkali-silica reaction, and chloride-induced corrosion [191].

The partial replacement of cement with RHA can lead to a significant reduction in the embodied carbon of concrete, as the production of RHA is less energy-intensive and has a lower carbon footprint compared to traditional cement manufacturing [189]. The incorporation of RHA in concrete can enhance the overall environmental performance of the material, including reductions in energy consumption, water usage, and waste generation [191]. The combination of RHA with other supplementary cementitious materials, such as fly ash and slag cement, can further optimize the

sustainability of concrete by achieving even greater reductions in the carbon footprint [187].

Palm oil fuel ash (POFA)

Palm Oil Fuel Ash (POFA) is a by-product of burning palm oil biomass in power plants, containing high levels of silica, along with alumina and iron oxide, which give it pozzolanic properties. The fine particle size and morphology of POFA play a critical role in its reactivity as an SCM. When incorporated into concrete, POFA can improve workability by reducing water demand and enhance long-term compressive strength through its pozzolanic reactions with calcium hydroxide, which densify the cement matrix. Additionally, POFA increases concrete's resistance to chemical attacks such as sulfate and chloride exposure, thereby improving durability [192, 193].

Bagasse ash (BA)

Bagasse Ash (BA) is another agricultural waste by-product, generated from the combustion of bagasse, the fibrous residue left after extracting sugarcane juice. With a high silica content, BA exhibits strong pozzolanic properties, particularly when finely ground. Its inclusion in concrete enhances workability, reduces water demand, and strengthens the concrete over time through pozzolanic reactions with calcium hydroxide, producing additional cementitious compounds. The use of BA improves concrete's durability against chemical attacks, such as sulfates and chlorides, prolonging the life of structures. BA also significantly lowers the carbon footprint of concrete by substituting energy-intensive cement production with agricultural waste [190, 194–196].

Wood waste ash

Wood Waste Ash (WWA), produced from the combustion of wood residues such as sawdust and wood chips, contains high silica levels alongside calcium and potassium oxides, which contribute to its pozzolanic activity. WWA's fine particle size improves the workability of concrete, making the mix more cohesive. Its pozzolanic reaction with calcium hydroxide strengthens concrete over time, particularly in long-term applications. Replacing part of the cement with WWA promotes sustainable construction practices and contributes to a circular economy by repurposing wood waste [197–200].

Bamboo leaf ash (BLF)

Bamboo Leaf Ash (BLA), derived from the combustion of bamboo leaves, has gained attention for its potential as an

SCM. Its high silica content contributes to its pozzolanic properties, allowing it to improve the mechanical performance of concrete. Studies show that replacing cement with BLA (typically 10–20%) enhances compressive strength, flexural strength, and bonding in concrete. BLA also increases resistance to sulfate attacks by reducing permeability, while improving the workability of concrete due to its fine particle size [201–206].

Corn cob ash

Corn Cob Ash (CCA) is another agricultural by-product with significant potential as an SCM. Rich in silica, CCA improves the compressive strength of concrete, especially when used at optimal replacement levels (10–20%). The pozzolanic activity of CCA enhances bonding, reduces porosity, and increases durability, particularly in resisting sulfate attacks and reducing permeability. Additionally, CCA's fine particle size improves the workability of concrete, making it easier to handle. By utilizing CCA, agricultural waste is repurposed, reducing landfill waste and promoting sustainability. Incorporating CCA in concrete also reduces carbon emissions by decreasing the reliance on Portland cement [207–209].

Metakaolin

Metakaolin is a calcined clay material, primarily composed of kaolinite, that serves as a pozzolanic additive in concrete. It is produced by heating kaolin clay to approximately 600–800 °C, which enhances its reactivity.

Metakaolin significantly improves the compressive strength of concrete, particularly at early ages. Studies have shown that replacing 10–20% of Portland cement with metakaolin can lead to substantial gains in strength [210, 211]. Concrete incorporating metakaolin exhibits enhanced durability against chemical attacks, such as sulfate and chloride exposure. The fine particles of metakaolin fill voids in the cement matrix, resulting in a denser microstructure [211]. The addition of metakaolin can improve the workability of concrete mixes. Its fine particle size contributes to better flow and handling characteristics, although high dosages may require adjustments in water content [212]. Using metakaolin as a partial cement replacement can lower the carbon footprint of concrete by reducing the amount of Portland cement used [210].

Ternary SCM systems

Ternary supplementary cementitious material (SCM) systems, which combine two different SCMs with Portland cement, are engineered to optimize both concrete

performance and sustainability by harnessing the unique and complementary characteristics of each material. Commonly used ternary combinations include Fly Ash-Slag-Portland Cement, Fly Ash-Silica Fume-Portland Cement, and Slag-Silica Fume-Portland Cement, each offering distinct synergistic effects that enhance mechanical properties, durability, and carbon reduction potential.

For example, the Fly Ash-Slag-Portland Cement system leverages the latent hydraulic activity of slag and the pozzolanic reactivity of fly ash. Slag improves early strength development and reduces permeability, while fly ash contributes to long-term strength gain and resistance to alkali-silica reactions. This synergy results in concrete with enhanced compressive strength and improved durability against chemical attack, with studies showing chloride ion permeability reductions by over 30% compared to binary systems [213, 214]. From a sustainability perspective, replacing up to 50% of Portland cement in such ternary blends can lower CO₂ emissions by approximately 40–50%, due to reduced clinker content and utilization of industrial by-products [215].

The Fly Ash-Silica Fume-Portland Cement ternary system combines the fineness and high pozzolanic activity of silica fume with the slower-reacting fly ash. Silica fume optimizes the microstructure by filling pores and refining the calcium silicate hydrate (C-S-H) gel, leading to a marked enhancement in early-age compressive strength—improvements up to 21% have been documented with 15% silica fume incorporation [213]. Fly ash extends long-term strength development and improves workability. The synergy between these materials results in reduced permeability and enhanced resistance to chloride penetration and sulfate attack, contributing to extended service life of structures [216]. Environmentally, this combination can achieve carbon reductions approaching 60% when Portland cement replacement levels reach 30–40%, underscoring substantial sustainability benefits [214].

Similarly, the Slag-Silica Fume-Portland Cement blend exploits the latent hydraulic properties of slag along with the ultra-fine particle size and pozzolanic reactivity of silica fume. This combination enhances densification of the cement matrix, improving modulus of elasticity and tensile strength, while significantly decreasing permeability and shrinkage [217]. The synergy manifests in improved durability under aggressive environmental conditions, including freeze–thaw cycles and chloride exposure. The partial replacement of Portland cement by up to 50% in this ternary system translates into a considerable carbon footprint reduction, estimated at 45–55%, mainly by substantial clinker savings and incorporation of waste-derived materials [218].

Ternary SCM systems capitalize on the complementary hydration kinetics and microstructural refinement provided

by multiple SCMs, which cannot be achieved by binary blends alone. These synergistic effects not only enhance mechanical and durability performance but also contribute to significant carbon emission reductions—often up to 60%—by substantially lowering clinker content while valorizing industrial by-products like fly ash, slag, and silica fume. The optimized combinations thus represent a practical and effective strategy for producing sustainable, high-performance concrete that meets stringent environmental and structural demands [214, 219, 220] (Tables 3 and 4).

Low carbon concrete mixtures

Mixture design plays a crucial role in reducing the carbon footprint of concrete by optimizing the proportions of materials used without compromising the mechanical and durability properties of the final product. The use of aggregate optimized gradation and performance-based mixture design are other techniques capable of reducing carbon footprint of concrete.

Optimization of aggregate blends for low-carbon concrete

Aggregates make up about 60–75% of concrete, which means that the characteristics, proportions, and packing of the aggregates significantly influence the performance of the concrete. By optimizing the blend of aggregates, it's possible to enhance concrete performance and reduce the amount of cementitious materials needed in the mix [221]. The impact of aggregate sizes on the structural characteristics of conventional concrete has been thoroughly researched, providing important insights into how concrete behaves under different loading conditions. Recently, there has been a growing interest in examining how aggregate sizes affect the structural properties of concrete, as researchers and practitioners aim to enhance the performance of this innovative construction material [222, 223]. On the other hand, arranging and distributing aggregates can be an effective criterion for optimizing concrete aggregate blends for various applications. Additionally, continuous power curves can also serve as a criterion for this optimization. The optimization process should prioritize several performance metrics, such as compressive strength, workability, and durability. Consistent testing and assessment of these properties are vital to ensure that the optimized blends comply with the standards for low-carbon concrete [221].

Optimizing aggregate blends is key to developing low-carbon concrete, as it improves performance while minimizing environmental impact. Employing advanced grading

Table 3 Summary of SCMs properties and their effect of the properties of concrete

Parameter	Silica fume	Fly ash	Slag cement	Agricultural ashes	Metakaolin [210, 211]
Chemical composition	SiO ₂ , C	SiO ₂ , Al ₂ O ₃ , Fe ₂ O ₃ , CaO	SiO ₂ , CaO, MgO, Al ₂ O ₃	SiO ₂ , CaO, Al ₂ O ₃ , MgO	SiO ₂ , Al ₂ O ₃
Fineness (m ² /kg)	15,000–30,000	300–500	400–600	15,000–30,000	12,000–20,000
Density (g/cm ³)	2.2	2.2–2.8	2.9–3.0	2.2–2.5	2.5–2.6
Workability	Decreased	Increased	Decreased	Decreased	Decreased
Setting time	Slightly accelerated	Delayed	Slightly delayed	Similar or delayed	Accelerated
Early age strength	Increased	Decreased	Similar or Increased	Decreased	Increased
Long-term strength	Increased	Increased	Increased	Increased	Increased
Chloride ion penetration	Significantly reduced	Reduced	Reduced	Reduced	Reduced
Sulfate attack resistance	Improved	Improved	Significantly Improved	Improved	Improved
Freeze–thaw resistance	Variable	Improved	Improved	Improved	Improved
Alkali–silica reaction (ASR)	Mitigated	Mitigated	Mitigated	Mitigated	Mitigated
Permeability	Significantly reduced	Reduced	Reduced	Reduced	Reduced
Carbon footprint reduction	Higher	Moderate	Higher	Higher	Moderate
Cost implications	Higher	Lower	Similar or higher	Lower	Higher
Shrinkage	Increased	Reduced	Similar or reduced	Reduced	Similar or reduced
Heat of hydration	Similar or reduced	Reduced	Similar or reduced	Reduced	Reduced

techniques and emphasizing sustainability can significantly enhance concrete formulations.

Optimizing performance-based mix design for low-carbon concrete

The optimization of mix design presents significant challenges, as the properties of concrete are influenced by both the chemical and physical characteristics of its constituent materials, along with their respective proportions [224]. Performance-based mix design for low-carbon concrete focuses on achieving desired mechanical and durability properties while minimizing the environmental impact, particularly in terms of carbon emissions. Unlike prescriptive mix design, which specifies exact material quantities, performance-based design allows flexibility in choosing materials as long as the required performance criteria are met. This approach encourages the use of supplementary cementitious materials (SCMs), recycled aggregates, and optimized aggregate gradation to reduce the cement content, a major source of carbon emissions.

An approach to reducing the global warming potential (GWP) of concrete involves the partial substitution of ordinary Portland cement (OPC) with SCMs. [225]. By tailoring the mix to meet specific performance benchmarks, such as

strength and durability over the structure's life cycle, designers can reduce the carbon footprint of concrete without sacrificing quality. This method enables more sustainable and resource-efficient concrete production by encouraging the use of innovative materials and technologies.

Concrete mix design methodologies can be broadly classified into three categories: (i) experimental approaches, (ii) robust design methods, and (iii) intelligent techniques. Experimental methods necessitate extensive testing, frequently employing a trial-and-error strategy, making them most suitable for optimizing a limited range of materials [226, 227]. To mitigate the need for costly and time-intensive experimentation, several problem-solving techniques have been developed based on robust experimental design principles. In these methodologies, experimental tests are structured to encompass a predefined range of proportions for each component, with the resulting data utilized to create best-fit plots for comparable materials [228–230].

To further reduce the necessity for trial mixing while maximizing the range of design parameters considered, innovative intelligent techniques based on artificial intelligence (AI) have recently been implemented in concrete mix design. These methodologies have demonstrated the capability to deliver reliable predictions when supplied with a suitable range of existing experimental data [231, 232].

Table 4 Summary of SCMs advantages, limitations, and carbon emission reduction

SCMs	Advantages	Limitations	PC replacement (%)	Concrete carbon footprint Reduction (%)	Estimated CO ₂ reduction
Silica Fume [183, 184]	Increased compressive strength Enhanced durability Reduced permeability	High cost Requires careful handling	5–10	5–15	30–80 kg of CO ₂ per cubic meter
Class F Fly Ash [183, 186]	Improved workability Reduced heat of hydration Enhanced durability	High Variability Requires longer curing time	15–40	10–30	50–150 kg of CO ₂ per cubic meter
Class C Fly Ash [185–187]			20–35	10–25	50–120 kg of CO ₂ per cubic meter
Slag Cement [187]	Improved durability Reduced permeability Low heat of hydration	Slow strength development Requires specific curing conditions	30–50	30–50	100–300 kg of CO ₂ per cubic meter
Rice Husk Ash [191]	Enhanced strength Reduced environmental impact Improved workability	High variability Requires fine grinding	10–30	10–30	540–150 kg of CO ₂ per cubic meter
Palm Oil Fuel Ash (POFA) [192, 193]	Improved strength and durability Cost-effective	High variability in chemical composition Limited availability	10–20	10–15	20–60 kg of CO ₂ per cubic meter
Bagasse Ash (BA) [196, 200]	Enhances durability Reduces environmental impact Cost-effective	Requires processing High variability	10–20	8–12	20–50 kg of CO ₂ per cubic meter
Wood Waste Ash [197–200]	Lightweight Improved workability Reducing waste	Limited availability High variability	10–20	12–18	20–75 kg of CO ₂ per cubic meter
Bamboo Leaf Ash (BLA) [201–206]	Enhanced strength Reducing waste Sustainable source	Limited availability High variability	10–20	8–15	20–60 kg of CO ₂ per cubic meter
Corn Cob Ash (CCA) [207–209]	Improves compressive strength Reducing waste Cost-effective	Limited availability High variability	10–20	10–12	20–50 kg of CO ₂ per cubic meter
Metakaolin [210, 211]	Increased compressive strength Enhanced durability Reduced permeability	Higher cost Requires careful handling	10–20	15–25	50–100 kg of CO ₂ per cubic meter

Use of artificial intelligence in mix optimization

The relationship between carbon emissions, SCMs, and concrete strength is intricate, as different SCMs have varying levels of embodied carbon, with materials like fly ash and GGBS typically exhibiting lower carbon footprints than Portland cement. While some SCMs may lead to slower initial strength gains, they often enhance long-term strength and durability, which can reduce lifecycle carbon emissions and extend service life. The optimal replacement percentage of SCMs is not uniform and depends on the specific type and desired properties of the concrete, necessitating careful optimization to balance strength, durability, and environmental impact. In designing concrete mixtures, the primary objective is often to achieve adequate compressive strength while also considering cost and carbon footprint,

which can be challenging with traditional trial-and-error methods. This has led to a growing interest in innovative approaches to material design, with recent advancements in civil engineering emphasizing the potential of artificial intelligence (AI) technologies for optimizing concrete mixture designs. Machine learning algorithms, especially artificial neural networks (ANNs), are gaining traction for their predictive accuracy in concrete applications, including strength prediction and environmental impact assessments, facilitating quick evaluations of carbon footprints during the design phase [233, 234]. Hybrid models that integrate genetic algorithms and neural networks have been developed to optimize concrete mixtures, significantly reducing embodied CO₂ while maintaining strength [235]. However, most studies focus on cost optimization rather than carbon emissions, and these models often lack reliable reverse

analysis capabilities to meet specific performance criteria, highlighting a notable limitation in their use [236].

Several recent studies have demonstrated the power of artificial intelligence (AI) and machine learning (ML) in optimizing recycled aggregate concrete (RAC) mix designs for enhanced sustainability and performance. Deng, Xiao, Duan, Li and Guan [237] developed an intelligent RAC mix design system integrating machine learning and optimization algorithms. Among four tested models, a genetic algorithm-optimized backpropagation neural network (GABP) delivered the highest accuracy in predicting compressive strength. Sensitivity analysis highlighted cement content, water-to-cement ratio, and recycled aggregate water absorption as critical factors. Incorporating a carbon emission model that considers both embodied emissions and service-life sequestration, the study showed optimized mixes could achieve up to 68.4% net CO₂ reduction while maintaining strengths above 50 MPa. Dual-objective optimization confirmed the feasibility of high RCA use in low- to mid-strength concrete without compromising performance. Mohammadi Golafshani, Kim, Behnood, Ngo and Kashani [238] compiled a dataset of 3519 RAC samples to evaluate the impact of SCMs on compressive strength. Using a variety of ML models—including Elastic Net, Support Vector Machines, Random Forest, and gradient boosting—they found that a stacking ensemble achieved the best prediction performance, outperforming Category Boosting by 4.2% in root mean squared error. Sensitivity analysis, using both real and simulated data, identified testing age and cement content as dominant strength predictors. The study also applied a multi-objective water cycle algorithm to optimize strength, cost, and CO₂ emissions, and developed a user-friendly interface for generating sustainable RAC mix designs.

Zhang, Pan, Chang, Wang, Liu, Jie, Ma, Shi, Guo, Xue and Wang [239] used ML to construct a predictive model and generate millions of optimized RAC mix designs across a range of strength grades. Their analysis using the Maximum Information Coefficient (MIC) revealed that RA content and water-to-cement ratio most significantly influence strength. A life cycle assessment (LCA) model was integrated to account for carbon emissions and transportation impacts. The study showed that optimized RAC mixes can substantially reduce emissions and that innovations in transportation could further enhance RA utilization, particularly in low-strength applications. Gao, Wang and Chu [240] proposed an advanced optimization framework using a generative model to map the high-dimensional mix design space into a simplified two-dimensional latent space. This approach enabled efficient multi-objective optimization of compressive strength, CO₂ emissions, and cost. A supporting explainable regression model ensured the feasibility of

AI-generated mix designs. Validation with a public dataset confirmed the framework's potential for broad adoption in sustainable concrete production.

Energy-efficient production process

In cement and concrete production, reducing energy consumption is vital for achieving low-carbon goals. Innovations in manufacturing processes can lead to more energy-efficient techniques, reducing emissions at each stage of production.

Reduced calcination temperatures

One of the most significant breakthroughs has been the development of alternative cements that require lower calcination temperatures during production. Traditional Portland cement requires temperatures of around 1450 °C to produce clinker, a process responsible for a large share of cement's carbon footprint. By reducing the temperature required to around 900 °C, alternative clinkers like belite-rich cements or LC3 drastically cut energy consumption and associated CO₂ emissions [241, 242].

Alternative energy sources

Shifting away from fossil fuels and adopting renewable energy sources in cement production can also help lower carbon emissions. Some plants have started utilizing biomass, waste-derived fuels, or solar thermal energy to power the kilns, significantly reducing reliance on coal or natural gas. Additionally, using waste heat recovery systems can improve energy efficiency by capturing and reusing heat generated during production [243, 244].

Process innovations

Technological advances in cement grinding, such as high-efficiency separators and vertical roller mills, reduce the energy required to grind raw materials and clinker into cement. This, combined with digitalization and automation of production lines, allows for real-time monitoring of energy consumption, optimizing the use of resources and further lowering the environmental footprint [245].

Economic feasibility

The economic feasibility analysis highlights that many ACMs, SCMs, and innovative clinker technologies offer promising cost advantages compared to traditional Portland

cement, particularly when local raw materials are readily available. For instance, materials like calcined clay, fly ash, slag, and agricultural ashes can significantly reduce overall costs while simultaneously lowering carbon emissions. Technologies such as LC3 and ternary SCM systems demonstrate high economic and environmental value when regional supply chains and production infrastructure support their implementation. However, the analysis also reveals several persistent challenges. The cost-effectiveness of many ACMs and SCMs is highly dependent on local availability, quality consistency, and processing requirements. Some alternatives, like magnesium-based systems and activated glassy cements, remain expensive due to complex production processes and limited raw material access. Additionally, technologies such as CO₂-cured cement and MOMS require integration with carbon capture systems and policy support (e.g., carbon pricing) to become economically viable at scale. Overall, while sustainable cement alternatives show strong potential, their broader adoption hinges on addressing logistical, technological, and policy-related barriers. (See Tables in Sects. "Alternative Cementitious Materials (ACMs)", "Supplementary materials" and "Innovative clinkers in concrete" for detailed comparisons.)

Alternative cementitious materials (ACMs)

Material	Economic feasibility highlights	Cost comparison to portland cement	Regional context considerations
Calcium-based ACMs	Using calcined clay as partial replacement can lower clinker content by up to 50%, reducing costs by up to 25%. Industrial symbiosis with cement kiln dust (CKD) adds value [246]	Potentially up to 25% cheaper than traditional OPC depending on local clay availability [246]	More feasible in regions with abundant kaolinic clays; value-added use of CKD enhances local feasibility [247]
Alkali-activated materials (AAMs)	Can reduce CO ₂ emissions by up to 85%. Cost-effective when precursor materials and activators are locally available [248]. Waste-derived activators can reduce costs by up to 40% [249]	Generally cheaper than OPC if activators and precursors are locally sourced; costs vary with processing [250, 251]	Feasibility depends on local availability of fly ash, slag, and activators; transportation emissions impact cost [252]
Magnesium-based ACMs	Eco-friendly with good mechanical performance; uses dolomite waste and reactive MgO. Costs higher due to MgO availability and processing	Often more expensive than OPC due to raw material and processing costs [253, 254]	Region-dependent niche applications where magnesium sources and processing facilities exist

Material	Economic feasibility highlights	Cost comparison to portland cement	Regional context considerations
Activated glassy cements (AGC)	Promising mechanical and durability properties; economic challenges due to raw material and processing costs; complexity in mixture design limits scale-up [255, 256]	Typically more expensive than OPC given current production and sourcing methods [256]	Economic viability hinges on efficient activation processes and reliable raw material supply in the region
CO ₂ -cured cement	Enhances mechanical properties and corrosion resistance; economic benefits linked to CO ₂ capture integration; savings depend on carbon taxation and energy costs	Can be cost-competitive if integrated with CO ₂ capture and favorable policy frameworks [257, 258]	Adoption sensitive to carbon pricing policies, energy costs, and integration capabilities at cement plants

Supplementary materials

Material	Economic feasibility highlights	Cost comparison to portland cement	Regional context considerations
Silica fume	Increases strength and durability, allowing up to 70% cement replacement. Its fine size can affect workability [259, 260]. Generally cost-effective with cost savings from reduced cement usage	Generally cheaper than OPC, but handling challenges moderate widespread [261]	Availability varies; some regions face supply constraints. Slump loss requires admixtures
Fly ash	Offers significant cost reduction by lowering cement use up to ~20%. Strength reduction beyond optimum replacement levels raises issues [262, 263]. Positive environmental and economic impact	Often less expensive than OPC; effective disposal of industrial by-product adds value [264]	Regional supply variations affect transport costs and adoption. Quality inconsistencies exist
Slag cement	Reduces carbon footprint and production cost; improves durability [265, 266]. Energy consumption may slightly increase or remain stable relative to OPC [265]	Typically cheaper than OPC; cost-effective for blends up to 40–60% replacement [267]	Steel industry by-product availability varies; logistical factors important [268]

Material	Economic feasibility highlights	Cost comparison to portland cement	Regional context considerations
Agricultural ashes	Provides low-cost SCM from waste residues, reducing cement demand and CO ₂ emissions [269, 270]. Strength and durability benefits vary with ash quality	Cheaper than OPC due to waste origin; processing costs influence cost-efficiency	Heterogeneity and regional supply influence adoption; requires standardization efforts [271]
Metakaolin	Enhances strength at 5–20% replacement; calcination costs affect price [272, 273]. Use with waste tire aggregates may reduce costs and emissions [274]	More expensive than some SCMs but reduces overall costs by improving performance [275]	Regional availability of kaolin and energy for calcination are key to feasibility
Ternary SCM systems	Combining SCMs (fly ash, slag, silica fume) reduces cement use by up to 70% [276]. Enables cost savings with durability benefits [277]	Economical overall due to combination of low-cost SCMs; some admixture cost offsets exist [278]	Optimization needed for local material sourcing, curing conditions, and mix design

Innovative clinkers in concrete

Technology	Economic feasibility highlights	Cost comparison to portland cement	Regional context considerations
Reactive Belite-rich Portland Cement (RBPC) clinkers	Lower firing temperatures decrease energy costs and CO ₂ emissions; improved durability reduces maintenance. Investments needed for kiln modifications are offset by energy savings	Potentially more cost-effective than OPC depending on fuel and raw material costs	Feasible where temperature-controllable kilns and alternative fuels (e.g., rice husk) are accessible [279]

Technology	Economic feasibility highlights	Cost comparison to portland cement	Regional context considerations
Belite-Ye'elimite-Ferrite (BYF) Binder Technology	Reduces CO ₂ emissions by 20–30% [280]. Costly aluminous raw materials pose economic challenges, but alpha prime belite doping reduces costs [281]	Competitive if aluminous materials sourced economically; synergistic additives improve value [282]	Successful adoption depends on raw materials and process adaptation; steel passivation important for reinforcement [283]
Carbonatable Calcium Silicate Clinkers (CCSC)	Produced from low-cost, abundant materials; scalable and adaptable to conventional plants; enables CO ₂ sequestration [284, 285]. Waste heat utilization enhances economic benefits [286]	Cost-effective due to use of industrial waste and lowered firing temperatures [287]	Transportation and local energy costs critical; industrial waste availability influences feasibility
Magnesium Oxides Derived from Magnesium Silicates (MOMS)	Reactivity and natural abundance support cost-effectiveness; integration with CO ₂ mineralization pathways promising [288, 289]. Processing costs and market adoption depend on resource access [290]	Moderate costs influenced by precursor processing; niche market with potential for expansion	Economic viability linked to efficient processing and integration with carbon capture technologies
Limestone Calcined Clay Cement (LC3)	Reduces clinker content by ~40–50%, cutting CO ₂ emissions by 35–45%; potential cost savings up to 25% over OPC by using abundant kaolin-itic clays and integrating with existing systems [291]	Cost-competitive and potentially cheaper than OPC, especially with local clay sources [292]	Proven feasibility in diverse regions such as Russia and Malawi; requires rotary calciner installation [293]

Specific changes in building codes and material standards

Performance-based criteria rather than prescriptive standards

Current standards, largely prescriptive and centered on traditional Portland cement, need to evolve towards more performance-based criteria. This shift would focus on measurable metrics such as compressive strength, durability, permeability, shrinkage, and corrosion resistance, irrespective of the cementitious material type used. For example, codes should allow the use of ACMs and AAMs if they meet or exceed defined performance benchmarks [294, 295].

Inclusion of innovative material categories and definitions

Building codes must explicitly include and define categories for ACMs, AAMs, and other innovative binders, clarifying their composition, characteristics, and appropriate applications. This includes defining Alkali-Activated Binders, Magnesium-Based Binders, and CO₂-Cured Cement within standards to remove ambiguity and enable proper evaluation and certification [294, 295].

Establishment of standardized testing protocols and quality control measures

To ensure uniformity and reliability, new standardized test methods must be developed, or existing ones adapted to assess:

- Reactivity, pozzolanic activity, and hydration kinetics of alternative materials.
- Long-term durability under relevant environmental exposures.
- Mechanical properties including compressive, tensile, and flexural strength.
- Performance under specific curing regimes, including CO₂ curing.
- Compatibility with admixtures and reinforcement materials.

Furthermore, stricter quality control protocols and certification processes are needed for sourcing and processing industrial by-products used in ACMs and SCMs to reduce variability affecting performance and safety [295].

Guidelines for mix design and application practices

Codes should include guidelines on (a) Optimal proportions and combinations of ternary or quaternary blends with SCMs to maximize synergy. (b) Adjustments in workability, setting times, and curing methods for these unconventional materials. (c) Performance monitoring over service life with defined acceptance criteria. (d) Special attention to mitigating issues like increased shrinkage or reduced early strength in some binders through additives or curing techniques [296, 297].

Safety and environmental impact assessments

New regulations should mandate assessment of (a) Environmental benefits including life-cycle carbon footprint reductions. (b) Health and safety considerations during production, handling, and application (e.g., alkali exposure in AAMs). (c) Waste utilization standards ensuring contaminants or toxic substances are within safe limits. This will facilitate sustainable sourcing and encourage industry adoption supported by regulatory approval [294, 295].

Compatibility with reinforcement and structural elements

Material standards must address (a) Interaction between innovative cements and steel reinforcement including corrosion resistance. (b) Passivation and durability in chloride-rich or aggressive environments. (c) Impact of altered pore structure and microstructure on bonding and long-term structural performance [220, 294].

Encouraging pilot projects and field demonstrations through code flexibility

Codes can incorporate provisions for pilot and demonstration projects using innovative materials with performance monitoring requirements. This flexibility aids gradual market introduction and evidence-based updates of standards and construction practices [294].

Training and knowledge transfer requirements

Regulatory frameworks should include requirements or incentives for training engineers, contractors, and product certifiers on the use and handling of innovative cementitious materials. This helps in overcoming lack of familiarity and builds confidence in new materials [295].

Conclusion

The development of Alternative Cementitious Materials (ACMs) and alternative clinkers represents a transformative shift in reducing the carbon footprint of concrete, one of the largest contributors to global greenhouse gas emissions. These materials—including calcium aluminate cements, calcium sulfoaluminate cements, magnesium-based cements, and alkali-activated binders—offer lower CO₂ emissions by utilizing innovative chemical compositions and production methods. Furthermore, incorporating industrial by-products such as fly ash, slag, and calcined clays not only enhances sustainability but also improves concrete's durability and resistance to chemical degradation.

Despite these promising advancements, several challenges hinder the widespread adoption of ACMs. Key concerns include long-term durability, cost-effectiveness, and the availability of alternative raw materials. Additionally, the lack of standardized testing methods and building codes presents regulatory barriers to large-scale implementation. Addressing these issues requires collaboration among researchers, industry stakeholders, and policymakers to develop comprehensive guidelines and best practices that ensure performance and reliability.

To accelerate low-carbon concrete implementation, upcoming studies should prioritize these key knowledge gaps:

- Optimize cement replacement ratios and alternative binder formulations to balance performance and sustainability.
- Advance energy-efficient manufacturing processes and carbon capture technologies to further reduce emissions.
- Promote circular economy principles by integrating recycled materials and minimizing waste in concrete production.
- Develop industry standards, certifications, and regulatory frameworks to support the commercialization of ACMs and alternative clinkers.
- Improve understanding of long-term mechanical behavior, durability under various environmental conditions, and the effects of different binder combinations to ensure reliable structural performance.
- Develop methods to enhance early strength and reduce setting times of sustainable mixes to meet construction speed requirements.
- Establish uniform standards and update codes to accommodate new low-carbon materials, enabling smoother regulatory approval and market adoption.
- Standardize LCA methodologies and integrate durability data to accurately assess and communicate environmental benefits.
- Ensure consistent quality in large-scale production and secure sustainable sourcing of raw materials to support industrial adoption.
- Advance AI and modelling tools to optimize mixes balancing performance, cost, and sustainability.

Emerging technologies, such as additive manufacturing, self-healing materials, and bio-based admixtures, offer additional opportunities to enhance the sustainability and longevity of concrete structures. Meanwhile, innovative business models and market incentives can accelerate the adoption of low-carbon cement products.

The transition to a more sustainable built environment relies on the collective efforts of governments, industry leaders, and the scientific community. By continuing to innovate, integrate sustainable technologies, and align policy frameworks with global climate goals, the construction industry can make meaningful strides toward carbon neutrality. With sustained research and investment, ACMs and alternative clinkers will play a critical role in shaping the future of eco-friendly concrete construction.

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Declarations

Conflicts of interest The authors declare that there are no conflicts of interest related to this work. Additionally, this study received no funding from sources that could influence its findings.

Ethical approval The authors affirm that this research complies with ethical standards. No human participants or animals were involved in the study. All data collection, analysis, and experimental procedures were conducted in accordance with relevant institutional and national guidelines.

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